# TEST REPORT
# EN IEC 62368-1

## Audio/video, information and communication technology equipment
## Part 1: Safety requirements

| | |
|---|---|
| Report reference No ..... : | RXM260507050-SF |
| Compiled by (+ signature) ..... : | Max Liu Max Liu |
| Approved by (+ signature) ..... : | Alex Xu |
| Date of issue ..... : | 2026-06-15 |
| Testing laboratory ..... : | Bay Area Compliance Laboratories Corp. (Kunshan) |
| Address ..... : | No. 248, Chenghu Road, Kunshan, Suzhou, Jiangsu, China |
| Testing location ..... : | As above |
| Applicant's name ..... : | Quanzhou STAYTRUE Electronic Technology Co., Ltd |
| Address..... : | Unit 1A-2, 1st Floor, Building 1, Fengshu Logistics Park, No. 920, Binhu South Road, Taiwan Investment Zone, Quanzhou, Fujian Province, China |
| Manufacturer's name..... : | Quanzhou STAYTRUE Electronic Technology Co., Ltd |
| Address..... : | Unit 1A-2, 1st Floor, Building 1, Fengshu Logistics Park, No. 920, Binhu South Road, Taiwan Investment Zone, Quanzhou, Fujian Province, China |
| Factory's name ..... : | Quanzhou STAYTRUE Electronic Technology Co., Ltd |
| Address..... : | Unit 1A-2, 1st Floor, Building 1, Fengshu Logistics Park, No. 920, Binhu South Road, Taiwan Investment Zone, Quanzhou, Fujian Province, China |
| Standard ..... : | EN IEC 62368-1:2024+A11:2024 |
| Test sample(s) received ..... : | 2026-05-07 |
| Test in period..... : | 2026-05-29 to 2026-06-02 |
| Type of test object ..... : | Module SDC-LE-01D |
| Trademark ..... : | SDC-LE-01D |
| Tested model..... : | SDC-LE-01D-4pin |
| Multiple Model ..... : | SDC-LE-01D-5pin |
| Manufacturer..... : | Quanzhou STAYTRUE Electronic Technology Co., Ltd |
| Rating……..: | Input: 3.3Vdc |

BAY AREA COMPLIANCE LABORATORY CORP. BACL COMPLIANCE AND VERIFICATION

**Copy of marking plate: (The marking plate artwork appended to this report may be only a draft)**




- The CE marking may be lower than 5.0mm and WEEE symbol should be at least 7.0mm in height.

| Test item particulars: | |
|---|---|
| **Product group .....:** | ☐ end product ☒ built-in component |
| **Classification of use by .....:** | ☐ Ordinary person ☐ Children likely present<br>☒ Instructed person<br>☒ Skilled person |
| **Supply connection.....:** | ☐ AC mains ☐ DC mains<br>☒ not mains connected:<br>☒ ES1 ☐ ES2 ☐ ES3 |
| **Supply tolerance .....:** | ☐ +10%/-10%<br>☐ +20%/-15%<br>☐ + %/ - %<br>☒ None: not directly connect to mains. |
| **Supply connection – type .....:** | ☐ pluggable equipment type A -<br>☐ non-detachable supply cord<br>☐ appliance coupler<br>☐ direct plug-in<br>☐ pluggable equipment type B -<br>☐ non-detachable supply cord<br>☐ appliance coupler<br>☐ permanent connection<br>☐ mating connector<br>☒ other: not directly connect to mains. |
| **Considered current rating of protective device.....:** | ☐ A;<br>Location: ☐ building ☐ equipment<br>☒ N/A |
| **Equipment mobility .....:** | ☐ movable ☐ hand-held ☐ transportable<br>☐ direct plug-in ☐ stationary ☒ for building-in<br>☐ wall/ceiling-mounted ☐ SRME/rack-mounted<br>☐ other: |
| **Overvoltage category (OVC) .....:** | ☐ OVC I ☐ OVC II ☐ OVC III<br>☐ OVC IV ☒ other: Not AC mains |
| **Class of equipment .....:** | ☐ Class I ☐ Class II<br>☐ Class II with functional earthing<br>☒ Class III ☐ Not classified ☐ |
| **Special installation location .....:** | ☒ N/A ☐ restricted access area<br>☐ outdoor location ☐ |
| **Pollution degree (PD) .....:** | ☐ PD 1 ☒ PD 2 ☐ PD 3 |
| **Manufacturer's specified $T_{ma}$.....:** | 105 °C ☐ Outdoor: minimum °C |
| **IP protection class .....:** | ☒ IPX0 ☐ IP___ |
| **Power systems .....:** | ☐ TN ☐ TT ☐ IT - $V_{L-L}$<br>☒ not AC mains |
| **Altitude during operation (m) .....:** | ☒ 2000 m or less ☐ m |
| **Altitude of test laboratory (m) .....:** | ☒ 2000 m or less ☐ m |
| **Mass of equipment (kg) .....:** | 0.003Kg |

| | |
|---|---|
| **POSSIBLE TEST CASE VERDICTS:** | |
| - test case does not apply to the test object..... | N (N/A) |
| - test object does meet the requirement..... | P (Pass) |
| - test object does not meet the requirement..... | F (Fail) |

General remarks:

"(see remark #)" refers to a remark appended to the report.

(see appended table)" refers to a table appended to the report.

Throughout this report a ☐comma/ ☒point is used as the decimal separator.

**General product information and other remarks:**

The product tested with model name SDC-LE-01D-4pin is Module SDC-LE-01D (for built-in) which powered by DC power supply with 3.3Vdc. All circuit considered to ES1 circuit.

Maximum recommended ambient (Tma): 105°C.

Model Numbers differences statement:

As Declaration Letter's description, this report covers following model numbers SDC-LE-01D-4pin, SDC-LE-01D-5pin. They are electrically identical with the PCB design, circuit diagram and components. The differences is model name and the number of pin, which will not affect the test results.

Test Statement:

The testing is performing on SDC-LE-01D-4pin as worst case to represent all other model.

Sample No.: RXM260507050-1, RXM260507050-2

| OVERVIEW OF ENERGY SOURCES AND SAFEGUARDS | | | | |
|---|---|---|---|---|
| **Clause** | **Possible Hazard** | | | |
| 5 | Electrically-caused injury | | | |
| Class and Energy Source (e.g. ES3: Primary circuit) | Body Part (e.g. Ordinary) | Safeguards | | |
| | | B | S | R |
| ES1: All circuits | Instructed person, Skilled person | N/A | N/A | N/A |
| 6 | Electrically-caused fire | | | |
| Class and Energy Source (e.g. PS2: 100 Watt circuit) | Material part (e.g. Printed board) | Safeguards | | |
| | | B | 1st S | 2nd S |
| PS3: Rate input: 3.3Vdc | PCB | See 6.3 | See 6.4.6 | N/A |
| 7 | Injury caused by hazardous substances | | | |
| Class and Energy Source (e.g. Ozone) | Body Part (e.g., Skilled) | Safeguards | | |
| | | B | S | R |
| N/A | N/A | N/A | N/A | N/A |
| 8 | Mechanically-caused injury | | | |
| Class and Energy Source (e.g. MS3: Plastic fan blades) | Body Part (e.g. Ordinary) | Safeguards | | |
| | | B | S | R |
| MS1:Sharp edges and Corners | Instructed person, Skilled person | N/A | N/A | N/A |
| MS1:Equipment mass <7kg | Instructed person, Skilled person | N/A | N/A | N/A |
| 9 | Thermal burn | | | |
| Class and Energy Source (e.g. TS1: Keyboard caps) | Body Part (e.g., Ordinary) | Safeguards | | |
| | | B | S | R |
| TS1: Internal components | Instructed person, Skilled person | N/A | N/A | See clause 9.3 |
| 10 | Radiation | | | |
| Class and Energy Source (e.g. RS1: PMP sound output) | Body Part (e.g., Ordinary) | Safeguards | | |
| | | B | S | R |
| N/A | N/A | N/A | N/A | N/A |
| Supplementary Information:<br>"B" – Basic Safeguard; "S" – Supplementary Safeguard; "R" – Reinforced Safeguard | | | | |

| ENERGY SOURCE DIAGRAM |
|---|
| **Optional**. Manufacturers are to provide the energy sources diagram identify declared energy sources and identifying the demarcations are between power sources. Recommend diagram be provided included in power supply and multipart systems.<br><br>Insert diagram below. Example diagram designs are; Block diagrams; image(s) with layered data; mechanical drawings |
| See “ENERGY SOURCE IDENTIFICATION AND CLASSIFICATION TABLE”<br><br>☒ **ES** ☒ **PS** ☒ **MS** ☒ **TS** ☐ **RS** |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| **4** | **GENERAL REQUIREMENTS** | | P |
| 4.1.1 | Acceptance of materials, components and subassemblies | Refer to summary of testing and appended table 4.1.2. | P |
| 4.1.2 | Use of components | Certified components are used in accordance with their ratings, certifications and they comply with applicable parts of this standard.<br>Components, for which no relevant IEC-standard exists, have been tested under the conditions occurring in the equipment, using applicable parts of IEC 62368-1. | P |
| 4.1.3 | Equipment design and construction | Equipment is adequately designed and constructed. | P |
| 4.1.4 | Specified ambient temperature for outdoor use (°C) : | Not for outdoor use. | N |
| 4.1.5 | Constructions and components not specifically covered | No such part. | N |
| 4.1.8 | Liquids, refrigerants and liquid filled components (LFCs) | No such components | N |
| 4.1.15 | Markings and instructions | (See Annex F) | P |
| 4.4.3 | Safeguard robustness | | P |
| 4.4.3.1 | General | | P |
| 4.4.3.2 | Steady force tests | The equipment is built-in type, the overall compliance shall be investigated in the complete end system/equipment. | N |
| 4.4.3.3 | Drop tests | | N |
| 4.4.3.4 | Impact tests | | N |
| 4.4.3.5 | Internal accessible safeguard tests | | N |
| 4.4.3.6 | Glass impact tests | | N |
| 4.4.3.7 | Glass fixation test | | N |
| | Glass impact test (1J) | | N |
| | Push/pull test (10 N) | | N |
| 4.4.3.8 | Thermoplastic material tests | | N |
| 4.4.3.9 | Air comprising a safeguard | | N |
| 4.4.3.10 | Accessibility, glass, safeguard effectiveness | | N |
| 4.4.4 | Displacement of a safeguard by an insulating liquid | | N |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| 4.4.5 | Safety interlocks | | N |
| **4.5** | **Explosion** | | N |
| 4.5.1 | General | The equipment is built-in type, the overall compliance shall be investigated in the complete end system/equipment. | N |
| 4.5.2 | No explosion during normal/abnormal operating conditions | | N |
| | No harm by explosion during single fault conditions | | N |
| **4.6** | | | N |
| | Fix conductors and conductive parts not to defeat a safeguard | | N |
| | Compliance is checked by test ..... : | | N |
| **4.7** | **Equipment for direct insertion into mains socket–outlets** | | N |
| 4.7.2 | Mains plug part complies with relevant standard ... : | | N |
| 4.7.3 | Torque (Nm) ..... : | | N |
| **4.8** | **Equipment containing coin or button cell batteries** | | N |
| 4.8.1 | General | | N |
| 4.8.2 | Instructional safeguard ..... : | | N |
| 4.8.3 | Coin or button cell battery compartment, door or cover construction | | N |
| | Open torque test | | N |
| 4.8.4.2 | Stress relief test | | N |
| 4.8.4.3 | Battery replacement test | | N |
| 4.8.4.4 | Drop test | | N |
| 4.8.4.5 | Impact test | | N |
| 4.8.4.6 | Crush test | | N |
| 4.8.5 | Compliance | | N |
| | 30N force test with test probe | | N |
| | 20N force test with test hook | | N |
| **4.9** | **Likelihood of fire or shock due to entry of conductive object** | | N |
| **4.10** | **Component requirements** | | N |
| 4.10.1 | Disconnect device | | N |
| 4.10.2 | Switches and relays | | N |
| 4.10.3 | Mains power supply cords | | N |
| 4.10.4 | Batteries and their protection circuits | | N |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |

| | | | |
|---|---|---|---|
| **5** | **ELECTRICALLY-CAUSED INJURY** | | P |
| **5.2** | **Classification and limits of electrical energy sources** | | P |
| 5.2.2 | ES1 and ES2 limits | See below | P |
| 5.2.2.2 | Steady-state voltage and current limits ..... : | (See appended table 5.2) | P |
| 5.2.2.3 | Capacitance limits ..... : | | N |
| 5.2.2.4 | Single pulse limits..... : | No single pulses generated. | N |
| 5.2.2.5 | Limits for repetitive pulses..... : | No repetitive pulses generated. | N |
| 5.2.2.6 | Ringing signals | Not used an analogue telephone network ringing signal in the equipment. | N |
| 5.2.2.7 | Audio signals | | N |
| **5.3** | **Protection against electrical energy sources** | | N |
| 5.3.1 | General Requirements for accessible parts to ordinary, instructed and skilled persons | Only the ES1 exist for the EUT, no need any safeguard for ES1. | N |
| 5.3.1 a) | ES2/ES3 circuits that are not ES2/ES3 mains | | N |
| 5.3.1 b) | Skilled persons not unintentional contact ES3 bare conductors | | N |
| 5.3.2.1 | Accessibility to electrical energy sources and safeguards | Only ES1 circuit can be accessible. | N |
| | Accessibility to outdoor equipment bare parts | | N |
| 5.3.2.2 | Contact requirements | | N |
| | Test with test probe from Annex V..... : | | — |
| 5.3.2.2 a) | Air gap – electric strength test potential (V)..... : | | N |
| 5.3.2.2 b) | Air gap – distance (mm) ..... : | | N |
| 5.3.2.3 | Compliance | | N |
| 5.3.2.4 | Terminals for connecting stripped wire | | N |
| **5.4** | **Insulation materials and requirements** | | P |
| 5.4.1.2 | Properties of insulating material | | N |
| 5.4.1.3 | Compliance | | N |
| | Non-hygroscopic materials | | N |
| 5.4.1.4 | Maximum operating temperature for insulating materials ..... : | (See appended table 5.4.1.4, 9.3, B.1.5, B.2.6) | P |
| 5.4.1.5 | Pollution degrees..... : | PD2 | — |
| 5.4.1.5.2 | Test for pollution degree 1 environment and for an insulating compound | | N |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| 5.4.1.5.3 | Thermal cycling test | | N |
| 5.4.1.6 | Insulation in transformers with varying dimensions | | N |
| 5.4.1.7 | Insulation in circuits generating starting pulses | | N |
| 5.4.1.8 | Determination of working voltage ..... : | (See appended table 5.4.1.8) | N |
| 5.4.1.9 | Insulating surfaces | | N |
| 5.4.1.10 | Thermoplastic parts on which conductive metallic parts are directly mounted | | N |
| 5.4.1.10.2 | Vicat test ..... : | | N |
| 5.4.1.10.3 | Ball pressure test..... : | | N |
| 5.4.2 | Clearances | Class III equipment, only ES1 | N |
| 5.4.2.1 | General requirements | | N |
| | Clearances in circuits connected to AC Mains, Alternative method | | N |
| 5.4.2.2 | Procedure 1 for determining clearance | | N |
| | Temporary overvoltage ..... : | | — |
| 5.4.2.3 | Procedure 2 for determining clearance | | N |
| 5.4.2.3.2.2 | AC mains transient voltages ..... : | | — |
| 5.4.2.3.2.3 | DC mains transient voltages ..... : | | — |
| 5.4.2.3.2.4 | External circuit transient voltages ..... : | | — |
| 5.4.2.3.2.5 | Transient voltage determined by measurement..... : | | — |
| 5.4.2.3.3 | Exceptions of determining required withstand voltage ..... : | | N |
| 5.4.2.3.4 | Determining clearances using required withstand voltage ..... : | | N |
| 5.4.2.4 | Determining the adequacy of a clearance using an electric strength test ..... : | | N |
| 5.4.2.5 | Multiplication factors for clearances and test voltages ..... : | | — |
| 5.4.2.6 | Clearance measurement..... : | | N |
| 5.4.3 | Creepage distances | Class III equipment, only ES1 | N |
| 5.4.3.1 | General | | N |
| 5.4.3.3 | Material group and CTI..... : | | — |
| 5.4.3.4 | Creepage distances measurement..... : | | N |
| 5.4.4 | Solid insulation | | N |
| 5.4.4.1 | General requirements | | N |
| 5.4.4.2 | Minimum distance through insulation ..... : | | N |

| EN IEC 62368-1 | | | |
|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| 5.4.4.3 | Insulating compound forming solid insulation | | N |
| 5.4.4.4 | Solid insulation in semiconductor devices | | N |
| 5.4.4.5 | Insulating compound forming cemented joints | | N |
| 5.4.4.6 | Thin sheet material | | N |
| 5.4.4.6.1 | General requirements | | N |
| 5.4.4.6.2 | Separable thin sheet material | | N |
| | Number of layers (pcs) ..... : | | N |
| 5.4.4.6.3 | Non-separable thin sheet material | | N |
| | Number of layers (pcs) ..... : | | N |
| 5.4.4.6.4 | Standard test procedure for non-separable thin sheet material ..... : | | N |
| 5.4.4.6.5 | Mandrel test | | N |
| 5.4.4.7 | Solid insulation in wound components | | N |
| 5.4.4.9 | Solid insulation at frequencies >30 kHz, $E_P$, $K_R$, $d$, $V_{PW}$ (V) ..... : | | N |
| | Alternative by electric strength test, tested voltage (V), $K_R$ ..... : | | N |
| 5.4.5 | Antenna terminal insulation | | N |
| 5.4.5.1 | General | | N |
| 5.4.5.2 | Voltage surge test | | N |
| 5.4.5.3 | Insulation resistance (MΩ)..... : | | N |
| | Electric strength test ..... : | (See appended table 5.4.9) | N |
| 5.4.6 | Insulation of internal wire as part of supplementary safeguard | | N |
| 5.4.7 | Tests for semiconductor components and for cemented joints | | N |
| 5.4.8 | Humidity conditioning | | N |
| | Relative humidity (%), temperature (°C), duration (h) ..... : | | — |
| 5.4.9 | Electric strength test | Class III equipment, only ES1 | N |
| 5.4.10 | Safeguards against transient voltages from external circuits | | N |
| 5.4.10.1 | Parts and circuits separated from external circuits | | N |
| 5.4.10.2 | Test methods | | N |
| 5.4.10.2.1 | General | | N |
| 5.4.10.2.2 | Impulse test ..... : | | N |
| 5.4.10.2.3 | Steady-state test..... : | | N |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| 5.4.10.3 | Verification for insulation breakdown..... : | | N |
| 5.4.11 | Separation between external circuits and earth | | N |
| 5.4.11.1 | Exceptions to separation between external circuits and earth | | N |
| 5.4.11.2 | Requirements | | N |
| | Surge suppressors bridge separation between external circuit and earth | | N |
| | Rated operating voltage $U_{op}$ (V)..... : | | — |
| | Nominal voltage $U_{peak}$ (V)..... : | | — |
| | Max increase due to variation $\Delta U_{sp}$ ..... : | | — |
| | Max increase due to ageing $\Delta U_{sa}$ ..... : | | — |
| 5.4.11.3 | Test method and compliance ..... : | | N |
| | Test voltage (V) of additional test..... : | | — |
| | Measured current (mA) of additional test ..... : | | N |
| 5.4.12 | Insulating liquid | | N |
| 5.4.12.1 | General requirements | | N |
| 5.4.12.2 | Electric strength of an insulating liquid ..... : | | N |
| 5.4.12.3 | Compatibility of an insulating liquid | | N |
| | Thermal classification of IEC 60085..... : | | — |
| 5.4.12.4 | Container for insulating liquid | | N |
| **5.5** | **Components as safeguards** | | N |
| 5.5.1 | General | | N |
| 5.5.2 | Capacitors and RC units | | N |
| 5.5.2.1 | General requirement | | N |
| 5.5.2.2 | Capacitor discharge after disconnection of a connector..... : | | N |
| 5.5.3 | Transformers | | N |
| 5.5.4 | Optocouplers | | N |
| 5.5.5 | Relays | | N |
| 5.5.6 | Resistors | | N |
| | Application type of resistors..... : | | — |
| 5.5.7 | Surge suppressors | | N |
| | GDT……: | | N |
| 5.5.8 | Insulation between the mains and an external circuit consisting of a coaxial cable ..... : | | N |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| | Insulation resistance (MΩ)..... : | | N |
| | Electric strength test..... : | | N |
| 5.5.9 | Safeguards for socket-outlets in outdoor equipment | | N |
| | RCD rated residual operating current (mA)..... : | | — |
| **5.6** | **Protective conductor** | | N |
| 5.6.2 | Requirements for protective conductors | Class III equipment, only ES1 | N |
| 5.6.2.1 | General requirements | | N |
| 5.6.2.2 | Colour of insulation | | N |
| 5.6.3 | Requirements for protective earthing conductors | | N |
| | Protective earthing conductor size ($mm^2$) ..... : | | — |
| | Protective earthing conductor serving as a reinforced safeguard | | N |
| | Protective earthing conductor serving as a double safeguard | | N |
| 5.6.4 | Requirements for protective bonding conductors | | N |
| 5.6.4.1 | Protective bonding conductors | | N |
| | Protective bonding conductor size ($mm^2$)..... : | | — |
| 5.6.4.2 | Protective current rating (A)..... : | | — |
| 5.6.5 | Terminals for protective conductors | | N |
| 5.6.5.1 | Terminal size for connecting protective earthing conductors (mm)..... : | | N |
| | Terminal size for connecting protective bonding conductors (mm)..... : | | N |
| | Relevant IEC standard ..... : | | N |
| 5.6.5.2 | Corrosion | | N |
| 5.6.6 | Resistance of the protective bonding system | | N |
| 5.6.6.1 | Requirements | | N |
| 5.6.6.2 | Test method..... : | | N |
| 5.6.6.3 | Resistance (Ω) or voltage drop..... : | | N |
| 5.6.7 | Reliable connection of a protective earthing conductor | | N |
| 5.6.8 | Functional earthing | | N |
| | Conductor size ($mm^2$)..... : | | N |
| | Class II with functional earthing marking ..... : | | N |
| | Appliance inlet cl & cr (mm)..... : | | N |
| **5.7** | **Prospective touch voltage, touch current and protective conductor current** | | N |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| 5.7.2 | Measuring devices and networks | Class III equipment, only ES1 | N |
| 5.7.2.1 | Measurement of touch current | | N |
| 5.7.2.2 | Measurement of voltage | | N |
| 5.7.3 | Equipment set-up, supply connections and earth connections | | N |
| 5.7.4 | Unearthed accessible parts ..... : | | N |
| 5.7.5 | Earthed accessible conductive parts ..... : | | N |
| 5.7.6 | Requirements when touch current exceeds ES2 limits | | N |
| | Protective conductor current (mA)..... : | | N |
| | Instructional Safeguard..... : | | N |
| 5.7.7 | Prospective touch voltage and touch current associated with external circuits | | N |
| 5.7.7.1 | Touch current from coaxial cables | | N |
| 5.7.7.2 | Prospective touch voltage and touch current associated with paired conductor cables | | N |
| 5.7.8 | Summation of touch currents from external circuits | | N |
| | a) Equipment connected to an earthed external circuit, current (mA) ..... : | | N |
| | b) Equipment connected to an unearthed external circuit, current (mA) ..... : | | N |
| **5.8** | **Backfeed safeguard in battery backed up supplies** | | N |
| | Mains terminal ES ..... : | (See appended table 5.8) | N |
| | Air gap (mm)..... : | (See appended table 5.4.2, 5.4.3) | N |
| **6** | **ELECTRICALLY- CAUSED FIRE** | | P |
| **6.2** | **Classification of power sources and potential ignition sources** | | P |
| 6.2.2 | Power source circuit classifications ..... : | (See appended table 6.2.2) | P |
| 6.2.3 | Classification of potential ignition sources | | P |
| 6.2.3.1 | Arcing PIS ..... : | | N |
| 6.2.3.2 | Resistive PIS ..... : | (See appended table 6.2.3.2) | P |
| **6.3** | **Safeguards against fire under normal operating and abnormal operating conditions** | | P |
| 6.3.1 | No ignition and attainable temperature value less than 90 % defined by ISO 871 or less than 300 ˚C for unknown materials ..... : | (See appended table 5.4.1.4, 9.3, B.1.5 and B.2.6) | P |
| | Combustible materials not inside a fire enclosure.... : | | N |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |

| | | | |
|---|---|---|---|
| **6.4** | **Safeguards against fire under single fault conditions** | | P |
| 6.4.1 | Safeguard method ..... : | | — |
| 6.4.2 | Reduction of the likelihood of ignition under single fault conditions in PS1 circuits | | N |
| 6.4.3 | Reduction of the likelihood of ignition under single fault conditions in PS2 and PS3 circuits | | N |
| 6.4.3.1 | Supplementary safeguards | | N |
| 6.4.3.2 | Single fault conditions ..... : | (See appended table B.4) | N |
| | Special conditions for temperature limited by fuse | | N |
| 6.4.4 | Control of fire spread in PS1 circuits | | N |
| 6.4.5 | Control of fire spread in PS2 circuits | | P |
| 6.4.5.2 | Supplementary safeguards | | P |
| 6.4.6 | Control of fire spread in PS3 circuits | All components mounted on min.V-1 PCB. In addition, the equipment is a building-in type and evaluation is to be made during the end system/equipment. | P |
| 6.4.7 | Separation of combustible materials from a PIS | | N |
| 6.4.7.2 | Separation by distance | | N |
| 6.4.7.3 | Separation by a fire barrier | | N |
| 6.4.8 | Fire enclosures and fire barriers | The equipment is built-in type, the overall compliance shall be investigated in the complete end system/equipment. | N |
| 6.4.8.2 | Fire enclosure and fire barrier material properties | | N |
| 6.4.8.2.1 | Requirements for a fire barrier | | N |
| 6.4.8.2.2 | Requirements for a fire enclosure | | N |
| 6.4.8.3 | Constructional requirements for a fire enclosure and a fire barrier | | N |
| 6.4.8.3.1 | Fire enclosure and fire barrier openings | | N |
| 6.4.8.3.2 | Fire barrier dimensions | | N |
| 6.4.8.3.3 | Top openings and properties | | N |
| | Openings dimensions (mm)..... : | | N |
| | Flammability tests for the top of a fire enclosure | | N |
| 6.4.8.3.4 | Bottom openings and properties | | N |
| | Openings dimensions (mm)..... : | | N |
| | Flammability tests for the bottom of a fire enclosure | | N |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| | Instructional Safeguard ..... : | | N |
| 6.4.8.3.5 | Side openings and properties | | N |
| | Openings dimensions (mm)..... : | | N |
| 6.4.8.3.6 | Integrity of a fire enclosure, condition met: a), b) or c)..... : | | N |
| 6.4.8.4 | Separation of a PIS from a fire enclosure and a fire barrier distance (mm) or flammability rating ..... : | | N |
| 6.4.9 | Flammability of insulating liquid | | N |
| | Auto ignition temperature (°C)..... : | | N |
| | Flashpoint temperature (°C) ..... : | | N |
| **6.5** | **Internal and external wiring** | | N |
| 6.5.1 | General requirements | | N |
| 6.5.2 | Requirements for interconnection to building wiring ..... : | | N |
| 6.5.3 | Internal wiring size ($mm^2$) for socket-outlets ..... : | | N |
| **6.6** | **Safeguards against fire due to the connection to additional equipment** | | N |

| | | | |
|---|---|---|---|
| **7** | **INJURY CAUSED BY HAZARDOUS SUBSTANCES** | | N |
| **7.2** | **Reduction of exposure to hazardous substances** | | N |
| **7.3** | **Ozone exposure** | | N |
| **7.4** | **Use of personal safeguards or personal protective equipment (PPE)** | | N |
| | Personal safeguards and instructions ..... : | | — |
| **7.5** | **Use of instructional safeguards and instructions** | | N |
| | Instructional safeguard (ISO 7010) ..... : | | — |

| | | | |
|---|---|---|---|
| **8** | **MECHANICALLY-CAUSED INJURY** | | P |
| **8.2** | **Mechanical energy source classifications** | | P |
| **8.3** | **Safeguards against mechanical energy sources** | | N |
| **8.4** | **Safeguards against parts with sharp edges and corners** | | N |
| 8.4.1 | Requirements | MS1 | N |
| | Instructional Safeguard ..... : | | N |
| 8.4.2 | Compliance criteria | | N |
| **8.5** | **Safeguards against moving parts** | | N |
| 8.5.1 | Requirements | | N |
| | Fingers, jewellery, clothing, hair, etc., contact with MS2 or MS3 parts | | N |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| | MS2 or MS3 part required to be accessible for the function of the equipment | | N |
| | A manually activated stopping device for moving MS3 | | N |
| | Moving MS3 parts only accessible to skilled person | | N |
| 8.5.2 | Instructional safeguard ..... : | | N |
| 8.5.4 | Special categories of equipment containing moving parts | | N |
| 8.5.4.1 | General | | N |
| 8.5.4.2 | Equipment containing work cells with MS3 parts | | N |
| 8.5.4.2.1 | Protection of persons in the work cell | | N |
| 8.5.4.2.2 | Access protection override | | N |
| 8.5.4.2.2.1 | Override system | | N |
| 8.5.4.2.2.2 | Visual indicator | | N |
| 8.5.4.2.3 | Emergency stop system | | N |
| | Maximum stopping distance from the point of activation (m) ..... : | | N |
| | Space between end point and nearest fixed mechanical part (mm)..... : | | N |
| 8.5.4.2.4 | Endurance requirements | | N |
| | Mechanical system subjected to 100 000 cycles of operation | | N |
| | - Mechanical function check and visual inspection | | N |
| | - Cable assembly..... : | | N |
| 8.5.4.3 | Equipment having electromechanical device for destruction of media | | N |
| 8.5.4.3.1 | Equipment safeguards | | N |
| 8.5.4.3.2 | Instructional safeguards against moving parts..... : | | N |
| 8.5.4.3.3 | Disconnection from the supply | | N |
| 8.5.4.3.4 | Cut type and test force (N) ..... : | | N |
| 8.5.4.3.5 | Compliance | | N |
| 8.5.5 | High pressure lamps | | N |
| | Explosion test ..... : | | N |
| 8.5.5.3 | Glass particles dimensions (mm)..... : | | N |
| **8.6** | **Stability of equipment** | | N |
| 8.6.1 | Requirements | | N |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| | Instructional safeguard for MS2 and MS3 television sets..... : | | N |
| 8.6.2 | Static stability | | N |
| 8.6.2.2 | Static stability test..... : | | N |
| 8.6.2.3 | Downward force test | | N |
| 8.6.3 | Relocation stability | | N |
| | Wheels diameter (mm)..... : | | — |
| | Tilt test | | N |
| 8.6.4 | Glass slide test | | N |
| 8.6.5 | Horizontal force test..... : | | N |
| **8.7** | **Equipment mounted to wall, ceiling or other structure** | | N |
| 8.7.1 | Requirements | | N |
| | Mount means type..... : | | N |
| 8.7.2 | Test methods | | N |
| | Test 1, additional downwards force (N) ..... : | | N |
| | Horizontal force to a wall or another structure | | N |
| | Test 2, number of attachment points and test force (N) ..... : | | N |
| | Test 3, nominal diameter (mm) and applied torque (Nm) ..... : | | N |
| **8.8** | **Handles strength** | | N |
| 8.8.1 | General | | N |
| 8.8.2 | Handle strength test | | N |
| | Number of handles ..... : | | — |
| | Weight applied (kg)..... : | | — |
| **8.9** | **Wheels or casters attachment requirements** | | N |
| 8.9.2 | Pull test | | N |
| **8.10** | **Carts, stands and similar carriers** | | N |
| 8.10.1 | General | | N |
| 8.10.2 | Marking and instructions ..... : | | N |
| 8.10.3 | Cart, stand or carrier loading test | | N |
| | Loading force applied (N)..... : | | N |
| 8.10.4 | Cart, stand or carrier impact test | | N |
| | Loading force applied (N) on each supporting surface ..... : | | N |
| 8.10.5 | Mechanical stability | | N |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| | Force applied (N) ..... : | | N |
| 8.10.6 | Thermoplastic temperature stability | | N |
| **8.11** | **Mounting means for slide-rail mounted equipment (SRME)** | | N |
| 8.11.1 | General | | N |
| 8.11.2 | Requirements | | N |
| | Instructional Safeguard ..... : | | N |
| 8.11.3 | Mechanical strength test | | N |
| 8.11.3.1 | Downward force test, force applied (N) ..... : | | N |
| 8.11.3.2 | Lateral push force test | | N |
| 8.11.3.3 | Integrity of slide rail end stops | | N |
| 8.11.4 | Compliance criteria | | N |
| **8.12** | **Telescoping or rod antennas** | | |
| | No sharp edges or points | | N |
| | Button/ball diameter (mm)..... : | | N |
| **9** | **THERMAL BURN INJURY** | | P |
| **9.2** | **Thermal energy source classifications** | | P |
| **9.3** | **Touch temperature limits** | | P |
| 9.3.1 | Touch temperatures of accessible parts .....: | The equipment is built-in type, the overall compliance shall be investigated in the complete end system/equipment. | P |
| 9.3.2 | Test method and compliance | | P |
| **9.4** | **Safeguards against thermal energy sources** | | N |
| **9.5** | **Requirements for safeguards** | | N |
| 9.5.1 | Equipment safeguard | | N |
| 9.5.2 | Instructional safeguard.....: | | N |
| **9.6** | **Requirements for wireless power transmitters** | | N |
| 9.6.1 | General | | N |
| 9.6.2 | Specification of the foreign objects | | N |
| 9.6.3 | Test method and compliance criteria .....: | | N |
| **10** | **RADIATION** | | N |
| **10.2** | **Radiation energy source classifications** | | N |
| 10.2.1 | General classification | | N |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| | Lasers ..... : | | — |
| | Lamps and lamp systems ..... : | | — |
| | Image projectors ..... : | | — |
| | X-Ray ..... : | | — |
| | Personal music player ..... : | | — |
| **10.3** | **Safeguards against laser radiation** | | N |
| | The standard(s) equipment containing laser(s) comply ..... : | | N |
| **10.4** | **Safeguards against optical radiation from lamps and lamp systems (including LED types)** | | N |
| 10.4.1 | General requirements | | N |
| | Instructional safeguard provided for accessible radiation level needs to exceed | | N |
| | Risk group marking and location ..... : | | N |
| | Information for safe operation and installation | | N |
| 10.4.2 | Requirements for equipment safeguards | | N |
| | UV radiation exposure ..... : | | N |
| 10.4.3 | Instructional safeguard ..... : | | N |
| **10.5** | **Safeguards against X-radiation** | | N |
| 10.5.1 | Requirements | | N |
| | Instructional safeguard for skilled persons ..... : | | N |
| 10.5.3 | Maximum radiation (pA/kg) ..... : | | N |
| **10.6** | **Safeguards against acoustic energy sources** | | N |
| 10.6.1 | General | | N |
| 10.6.2 | Classification | | N |
| | Acoustic output $L_{Aeq,T}$, dB(A) ..... : | | N |
| | Unweighted RMS output voltage (mV) ..... : | | N |
| | Digital output signal (dBFS) ..... : | | N |
| 10.6.3 | Requirements for dose-based systems | | N |
| 10.6.3.1 | General requirements | | N |
| 10.6.3.2 | Dose-based warning and automatic decrease | | N |
| 10.6.3.3 | Exposure-based warning and requirements | | N |
| | 30 s integrated exposure level (MEL30) ..... : | | N |
| | Warning for MEL ≥ 100 dB(A) ..... : | | N |
| 10.6.4 | Measurement methods | | N |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| 10.6.5 | Protection of persons | | N |
| | Instructional safeguards.....: | | N |
| 10.6.6 | Requirements for listening devices (headphones, earphones, etc.) | | N |
| 10.6.6.1 | Corded listening devices with analogue input | | N |
| | Listening device input voltage (mV).....: | | N |
| 10.6.6.2 | Corded listening devices with digital input | | N |
| | Max. acoustic output $L_{Aeq,T}$, dB(A).....: | | N |
| 10.6.6.3 | Cordless listening devices | | N |
| | Max. acoustic output $L_{Aeq,T}$, dB(A).....: | | N |
| **B** | **NORMAL OPERATING CONDITION TESTS, ABNORMAL OPERATING CONDITION TESTS AND SINGLE FAULT CONDITION TESTS** | | P |
| **B.1** | **General** | | P |
| B.1.5 | Temperature measurement conditions | (See appended table 5.4.1.4, 9.3, B.1.5, B.2.6) | P |
| B.1.6 | Specific output conditions | | N |
| **B.2** | **Normal operating conditions** | | P |
| B.2.1 | General requirements .....: | (See Test Item Particulars and appended test tables) | P |
| | Audio Amplifiers and equipment containing an audio amplifiers.....: | | N |
| B.2.3 | Supply voltage and tolerances | | N |
| B.2.5 | Input test .....: | (See appended table B.2.5) | P |
| B.2.6.4 | Equipment intended for building-in or rack-mounting | | P |
| **B.3** | **Simulated abnormal operating conditions** | | N |
| B.3.1 | General | | N |
| B.3.2 | Covering of ventilation openings | | N |
| | Instructional safeguard.....: | | N |
| B.3.3 | DC mains polarity test | | N |
| B.3.4 | Setting of voltage selector | | N |
| B.3.5 | Maximum load at output terminals | | N |
| B.3.6 | Reverse battery polarity | | N |
| B.3.7 | Audio amplifier abnormal operating conditions | | N |
| B.3.8 | Safeguards functional during and after abnormal operating conditions.....: | | N |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |

| | | | |
|---|---|---|---|
| **B.4** | **Simulated single fault conditions** | | P |
| B.4.1 | General | | P |
| B.4.2 | Temperature controlling device | No such parts used for the equipment. | N |
| B.4.3 | Blocked motor test | | N |
| B.4.4 | Functional insulation | See below | P |
| B.4.4.1 | Short circuit of clearances for functional insulation | (See appended table B.4) | P |
| B.4.4.2 | Short circuit of creepage distances for functional insulation | (See appended table B.4) | P |
| B.4.4.3 | Short circuit of functional insulation on coated printed boards | | N |
| B.4.5 | Short-circuit and interruption of electrodes in tubes and semiconductors | (See appended table B.4) | P |
| B.4.6 | Short circuit or disconnection of passive components | (See appended table B.4) | P |
| B.4.7 | Continuous operation of components | | N |
| B.4.8 | Compliance criteria during and after single fault conditions.....: | (See appended table B.4)<br>During and after single fault conditions, accessible parts does not exceed the relevant energy class and no flame and ignition inside the equipment. | P |
| B.4.9 | Battery charging and discharging under single fault conditions | | N |
| **C** | **UV RADIATION** | | N |
| **C.1** | **Protection of materials in equipment from UV radiation** | | N |
| C.1.2 | Requirements | | N |
| C.1.3 | Test method and compliance criteria | | N |
| **C.2** | **UV light conditioning test** | | N |
| C.2.1 | Test apparatus.....: | | N |
| C.2.2 | Mounting of test samples | | N |
| C.2.3 | Carbon-arc light-exposure test | | N |
| C.2.4 | Xenon-arc light-exposure test | | N |
| **D** | **TEST GENERATORS** | | N |
| **D.1** | **Impulse test generators** | | N |
| **D.2** | **Antenna interface test generator** | | N |
| **D.3** | **Electronic pulse generator** | | N |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| **E** | **TEST CONDITIONS FOR EQUIPMENT INTENDED TO AMPLIFY AUDIO SIGNALS** | | N |
| **E.1** | **Electrical energy source classification for audio signals** | | N |
| | Maximum non-clipped output power (W) ..... : | | — |
| | Rated load impedance (Ω) ..... : | | — |
| | Open-circuit output voltage (V) ..... : | | — |
| | Instructional safeguard..... : | | — |
| **E.2** | **Audio signals used during test** | | N |
| E.2.1 | Pink noise test signal | | N |
| E.2.2 | Sine-wave signal | | N |
| **E.3** | **Operating conditions of equipment containing an audio amplifier** | | N |
| E.3.1 | Normal operating conditions | | N |
| E.3.2 | Abnormal operating conditions | | N |
| E.3.3 | Audio equipment temperature measurement conditions..... : | | N |
| **F** | **EQUIPMENT MARKINGS, INSTRUCTIONS, AND INSTRUCTIONAL SAFEGUARDS** | | P |
| **F.1** | **General** | | P |
| | Language ..... : | English verified.<br>A language accepted in the respective countries. | — |
| **F.2** | **Letter symbols and graphical symbols** | | P |
| F.2.1 | Letter symbols according to IEC 60027-1 | Letter symbols for quantities and units are complied with<br>IEC 60027-1. | P |
| F.2.2 | Graphic symbols according to IEC, ISO or manufacturer specific | Graphical symbols are complied with IEC 60417, ISO 3864-2, ISO7000 or ISO 7010. | P |
| **F.3** | **Equipment markings** | | P |
| F.3.1 | Equipment marking locations | | P |
| F.3.2 | Equipment identification markings | See below | P |
| F.3.2.1 | Manufacturer identification ..... : | See copy of marking plate | P |
| F.3.2.2 | Model identification ..... : | See copy of marking plate | P |
| F.3.3 | Equipment rating markings | | N |
| F.3.3.1 | Equipment with direct connection to mains | Not directly connected to the mains | N |
| F.3.3.2 | Equipment without direct connection to mains | Not directly connected to the mains | P |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| F.3.3.3 | Nature of the supply voltage ..... : | DC | N |
| F.3.3.4 | Rated voltage ..... : | | N |
| F.3.3.5 | Rated frequency ..... : | | N |
| F.3.3.6 | Rated current or rated power ..... : | | N |
| F.3.3.7 | Equipment with multiple supply connections | | N |
| F.3.4 | Voltage setting device | No voltage setting device | N |
| F.3.5 | Markings on terminals and operating devices | | N |
| F.3.5.1 | Mains appliance outlet and socket-outlet markings : | | N |
| F.3.5.2 | Switch position identification marking ..... : | | N |
| F.3.5.3 | Replacement fuse identification and rating markings ..... : | | N |
| | Instructional safeguards for neutral fuse ..... : | | N |
| F.3.5.4 | Replacement battery identification marking ..... : | | N |
| F.3.5.5 | Neutral conductor terminal | | N |
| F.3.5.6 | Terminal marking location | | N |
| F.3.6 | Equipment markings related to equipment classification | | N |
| F.3.6.1 | Class I equipment | | N |
| F.3.6.1.1 | Protective earthing conductor terminal ..... : | | N |
| F.3.6.1.2 | Protective bonding conductor terminals ..... : | | N |
| F.3.6.2 | Equipment class marking ..... : | | N |
| F.3.6.3 | Functional earthing terminal marking ..... : | | N |
| F.3.7 | Equipment IP rating marking ..... : | IPX0 | N |
| F.3.8 | External power supply unit output marking ..... : | | N |
| F.3.9 | Durability, legibility and permanence of markings | The markings on the equipment is durable and legible, and easily discernable under normal lighting conditions | P |
| F.3.10 | Test for permanence of markings | Rubbing the marking by hand for 15 s with piece of cloth soaked with water and, at a different place for 15 s with a piece of cloth soaked with petroleum spirit. After this test, marking is legible and cannot be easily possible to remove marking and show no curling | P |
| **F.4** | **Instructions** | | P |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| | – Information prior to installation and initial use | | P |
| | – Equipment for use in locations where children not likely to be present | | N |
| | – Instructions for installation and interconnection | | P |
| | – Equipment intended for use only in restricted access area | | N |
| | – Equipment intended to be fastened in place | | P |
| | – Instructions for audio equipment terminals | | N |
| | – Protective earthing used as a safeguard | | N |
| | – Protective conductor current exceeding ES2 limits | | N |
| | – Graphic symbols used on equipment | | N |
| | – Permanently connected equipment not provided with all-pole mains switch | | N |
| | – Replaceable components or modules providing safeguard function | | N |
| | – Equipment containing insulating liquid | | N |
| | – Installation instructions for outdoor equipment | | N |
| **F.5** | Instructional safeguards | | P |
| **G** | **COMPONENTS** | | **N** |
| **G.1** | **Switches** | | N |
| G.1.1 | General | | N |
| G.1.2 | Ratings, endurance, spacing, maximum load | | N |
| G.1.3 | Test method and compliance criteria | | N |
| **G.2** | **Relays** | | N |
| G.2.1 | Requirements and compliance criteria | | N |
| G.2.2 | Overload test | | N |
| G.2.3 | Relay controlling connectors supplying power to other equipment | | N |
| G.2.4 | Test method and compliance criteria | | N |
| **G.3** | **Protective devices** | | N |
| G.3.1 | Thermal cut-offs | | N |
| | Thermal cut-outs separately approved according to IEC 60730-1 with conditions indicated in a) & b) | | N |
| | Thermal cut-outs tested as part of the equipment as indicated in c) | | N |
| G.3.1.2 | Test method and compliance criteria | | N |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| G.3.2 | Thermal links | | N |
| G.3.2.1 | a) Thermal links tested separately according to IEC 60691 with specifics | | N |
| | b) Thermal links tested as part of the equipment | | N |
| G.3.2.2 | Test method and compliance criteria | | N |
| G.3.3 | PTC thermistors | | N |
| G.3.4 | Overcurrent protection devices | | N |
| G.3.5 | Safeguards components not mentioned in G.3.1 to G.3.4 | | N |
| G.3.5.1 | Non-resettable devices suitably rated and marking provided | | N |
| G.3.5.2 | Single faults conditions .....: | | N |
| **G.4** | **Connectors** | | N |
| G.4.1 | Spacings | No such component. | N |
| G.4.2 | Mains connectors configuration .....: | | N |
| G.4.3 | Plug is shaped that insertion into mains socket-outlets or appliance coupler is unlikely | | N |
| **G.5** | **Wound components** | | N |
| G.5.1 | Wire insulation in wound components | No such component. | N |
| G.5.1.2 | Protection against mechanical stress | | N |
| G.5.2 | Endurance test | | N |
| G.5.2.1 | General test requirements | | N |
| G.5.2.2 | Heat run test | | N |
| | Test time (days per cycle).....: | | — |
| | Test temperature (°C) .....: | | — |
| G.5.2.3 | Wound components supplied from the mains | | N |
| G.5.2.4 | Compliance criteria | | N |
| G.5.3 | Transformers | | N |
| G.5.3.1 | General | | N |
| | Compliance method .....: | | N |
| G.5.3.2 | Insulation | | N |
| | Protection from displacement of windings .....: | | — |
| G.5.3.3 | Transformer overload tests | | N |
| G.5.3.3.1 | Test conditions | | N |
| | Position .....: | | N |
| | Method of protection .....: | | N |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| G.5.3.3.2 | Winding temperatures | | N |
| G.5.3.3.3 | Winding temperatures - alternative test method | | N |
| G.5.3.4 | Transformers using FIW | | N |
| G.5.3.4.1 | General | | N |
| | FIW wire nominal diameter ..... : | | — |
| G.5.3.4.2 | Transformers with basic insulation only | | N |
| G.5.3.4.3 | Transformers with double insulation or reinforced insulation ..... : | | N |
| G.5.3.4.4 | Transformers with FIW wound on metal or ferrite core | | N |
| G.5.3.4.5 | Thermal cycling test and compliance | | N |
| G.5.3.4.6 | Partial discharge test | | N |
| G.5.4 | Motors | | N |
| G.5.4.1 | General requirements | No such component. | N |
| G.5.4.2 | Motor overload test conditions | | N |
| G.5.4.3 | Running overload test | | N |
| G.5.4.4 | Locked-rotor overload test | | N |
| | Test duration (days) ..... : | | — |
| | Electric strength test ..... : | | N |
| G.5.4.5 | Running overload test for DC motors | | N |
| G.5.4.5.2 | Tested in the unit | | N |
| | Electric strength test ..... : | | N |
| G.5.4.5.3 | Alternative method | | N |
| | Electric strength test ..... : | | N |
| G.5.4.6 | Locked-rotor overload test for DC motors | | N |
| G.5.4.6.2 | Tested in the unit | | N |
| | Maximum Temperature (°C) ..... : | | N |
| | Electric strength test ..... : | | N |
| G.5.4.6.3 | Alternative method | | N |
| | Electric strength test ..... : | | N |
| G.5.4.7 | Motors with capacitors | | N |
| G.5.4.8 | Three-phase motors | | N |
| G.5.4.9 | Series motors | | N |
| | Operating voltage (V) ..... : | | — |
| **G.6** | **Wire Insulation** | | N |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| G.6.1 | General | | N |
| G.6.2 | Enamelled winding wire insulation | | N |
| **G.7** | **Mains power supply cords and interconnection cables** | | N |
| G.7.1 | General requirements | | N |
| | Type ..... : | | — |
| G.7.2 | Cross sectional area (mm$^2$ or AWG) ..... : | | N |
| G.7.3 | Cord anchorages and strain relief | | N |
| G.7.3.2 | Cord strain relief | | N |
| G.7.3.2.1 | Requirements | | N |
| | Strain relief test force (N) ..... : | | N |
| G.7.3.2.2 | Strain relief mechanism failure | | N |
| G.7.3.2.3 | Cord sheath or jacket position, distance (mm) ..... : | | N |
| G.7.3.2.4 | Strain relief and cord anchorage material | | N |
| G.7.4 | Cord Entry | | N |
| G.7.5 | Non-detachable cord bend protection | | N |
| G.7.5.1 | Requirements | | N |
| G.7.5.2 | Test method and compliance criteria | | N |
| | Overall diameter or minor overall dimension, *D* (mm) ..... : | | — |
| | Radius of curvature after test (mm) ..... : | | — |
| G.7.6 | Supply wiring space | | N |
| G.7.6.1 | General requirements | | N |
| G.7.6.2 | Stranded wire | | N |
| G.7.6.2.1 | Requirements | | N |
| G.7.6.2.2 | Test with 8 mm strand | | N |
| **G.8** | **Varistors** | | N |
| G.8.1 | General requirements | | N |
| G.8.2 | Safeguards against fire | | N |
| G.8.2.1 | General | | N |
| G.8.2.2 | Varistor overload test | | N |
| G.8.2.3 | Temporary overvoltage test | | N |
| **G.9** | **Integrated circuit (IC) current limiters** | | N |
| G.9.1 | Requirements | No such component. | N |
| | IC limiter output current (max. 5A) ..... : | | — |
| | Manufacturers' defined drift ..... : | | — |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| G.9.2 | Test Program | | N |
| G.9.3 | Compliance criteria | | N |
| **G.10** | **Resistors** | | N |
| G.10.1 | General | No such component. | N |
| G.10.2 | Conditioning | | N |
| G.10.3 | Resistor test | | N |
| | Changes of resistance (%) .....: | | N |
| | Measured current with the lowest resistance value : | | N |
| G.10.4 | Voltage surge test | | N |
| | Changes of resistance (%) .....: | | N |
| G.10.5 | Impulse test | | N |
| | Changes of resistance (%) .....: | | N |
| G.10.6 | Overload test | | N |
| | Changes of resistance (%) .....: | | N |
| **G.11** | **Capacitors and RC units** | | N |
| G.11.1 | General requirements | No such component. | N |
| G.11.2 | Conditioning of capacitors and RC units | | N |
| G.11.3 | Rules for selecting capacitors | | N |
| **G.12** | **Optocouplers** | | N |
| | Optocouplers comply with IEC 60747-5-5 with specifics | | N |
| | Type test voltage $V_{ini,a}$ .....: | | — |
| | Routine test voltage, $V_{ini,b}$ .....: | | — |
| **G.13** | **Printed boards** | | N |
| G.13.1 | General requirements | | N |
| G.13.2 | Uncoated printed boards | | N |
| G.13.3 | Coated printed boards | | N |
| G.13.4 | Insulation between conductors on the same inner surface | | N |
| G.13.5 | Insulation between conductors on different surfaces | | N |
| | Distance through insulation .....: | | N |
| | Number of insulation layers (pcs) .....: | | — |
| G.13.6 | Tests on coated printed boards | | N |
| G.13.6.1 | Sample preparation and preliminary inspection | | N |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| G.13.6.2 | Test method and compliance criteria | | N |
| **G.14** | **Coating on components terminals** | | N |
| G.14.1 | Requirements ..... : | (See Clause G.13) | N |
| **G.15** | **Pressurized liquid filled components or LFC assemblies** | | N |
| G.15.1 | Requirements | No such component. | N |
| G.15.2 | Test methods and compliance criteria for self-contained LFC | | N |
| G.15.2.1 | Hydrostatic pressure test, applied test pressure .... : | | N |
| G.15.2.2 | Creep resistance test | | N |
| G.15.2.3 | Tubing and fittings compatibility test, the change of tensile strength (%) ..... : | | N |
| G.15.2.4 | Vibration test | | N |
| G.15.2.5 | Thermal cycling test, test temperature (°C) ..... : | | N |
| G.15.2.6 | Force test | | N |
| G.15.2.7 | Compliance criteria | | N |
| G.15.3 | Test methods and compliance for a modular LFC | | N |
| G.15.3.2 | Hydrostatic pressure test, applied test pressure .... : | | N |
| G.15.3.3 | Creep resistance test | | N |
| G.15.3.4 | Tubing and fittings compatibility test, the change of tensile strength (%) ..... : | | N |
| G.15.3.5 | Thermal cycle test, test temperature (°C) ..... : | | N |
| G.15.3.6 | Force test | | N |
| G.15.3.7 | Compliance criteria | | N |
| **G.16** | **IC including capacitor discharge function (ICX)** | | N |
| G.16.1 | Condition for fault tested is not required | No such component. | N |
| | ICX with associated circuitry tested in equipment | | N |
| | ICX tested separately | | N |
| G.16.2 | Tests | | N |
| | Smallest capacitance and smallest resistance specified by ICX manufacturer for impulse test ..... : | | — |
| | Mains voltage that impulses to be superimposed on ..... : | | — |
| | Largest capacitance and smallest resistance for ICX tested by itself for 10000 cycles test..... : | | — |
| G.16.3 | Capacitor discharge test ..... : | (See appended table 5.5.2.2) | N |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |

| | | | |
|---|---|---|---|
| **H** | **CRITERIA FOR TELEPHONE RINGING SIGNALS** | | **N** |
| **H.1** | **General** | | N |
| **H.2** | **Method A** | | N |
| **H.3** | **Method B** | | N |
| H.3.1 | Ringing signal | | N |
| H.3.1.1 | Frequency (Hz) .....: | | — |
| H.3.1.2 | Voltage (V) .....: | | — |
| H.3.1.3 | Cadence; time (s) and voltage (V) .....: | | — |
| H.3.1.4 | Single fault condition current (mA):.....: | | — |
| H.3.2 | Tripping device and monitoring voltage | | N |
| H.3.2.1 | Conditions for use of a tripping device or a monitoring voltage | | N |
| H.3.2.2 | Tripping device | | N |
| H.3.2.3 | Monitoring voltage (V).....: | | N |
| **J** | **INSULATED WINDING WIRES FOR USE WITHOUT INTERLEAVED INSULATION** | | **N** |
| **J.1** | **General** | | N |
| | Winding wire insulation .....: | | — |
| | Solid round winding wire, diameter (mm) .....: | | N |
| | Solid square and rectangular (flatwise bending) winding wire, cross-sectional area ($mm^2$).....: | | N |
| **J.2/J.3** | Tests and Manufacturing | (See separate test report) | — |
| **K** | **SAFETY INTERLOCKS** | | N |
| **K.1** | **General requirements** | | N |
| | Instructional safeguard.....: | | N |
| **K.2** | **Components of safety interlock safeguard mechanism** | | N |
| **K.3** | **Inadvertent change of operating mode** | | N |
| **K.4** | **Interlock safeguard override** | | N |
| **K.5** | **Fail-safe** | | N |
| K.5.1 | Under single fault condition | | N |
| **K.6** | **Mechanically operated safety interlocks** | | N |
| K.6.1 | Endurance requirement | | N |
| **K.7** | **Interlock circuit isolation** | | N |
| K.7.1 | Separation distance for contact gaps & interlock circuit elements | | N |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| | In circuit connected to mains, separation distance for contact gaps (mm) ..... : | | N |
| | In circuit isolated from mains, separation distance for contact gaps (mm) ..... : | | N |
| | Electric strength test before and after the test of K.7.2 ..... : | | N |
| K.7.2 | Overload test, Current (A) ..... : | | N |
| K.7.3 | Endurance test | | N |
| K.7.4 | Electric strength test | | N |
| **L** | **DISCONNECT DEVICES** | | N |
| **L.1** | **General requirements** | | N |
| **L.2** | **Permanently connected equipment** | | N |
| | Instructions for permanently connected equipment | | N |
| **L.3** | **Parts that remain energized** | | N |
| **L.4** | **Single-phase equipment** | | N |
| | Instructions for single pole disconnect device | | N |
| **L.5** | **Three-phase equipment** | | N |
| **L.6** | **Switches as disconnect devices** | | N |
| **L.7** | **Plugs as disconnect devices** | | N |
| | Instructions for pluggable equipment | | — |
| **L.8** | **Multiple power sources** | | N |
| | Instructional safeguard ..... : | | N |
| **M** | **EQUIPMENT CONTAINING BATTERIES AND THEIR PROTECTION CIRCUITS** | | N |
| **M.1** | **General requirements** | | N |
| **M.2** | **Safety of batteries and their cells** | | N |
| M.2.1 | Batteries and their cells comply with relevant IEC standards ..... : | | N |
| **M.3** | **Protection circuits for batteries provided within the equipment** | | N |
| M.3.1 | Requirements | | N |
| M.3.2 | Test method | | N |
| | Overcharging of a rechargeable battery | | N |
| | Excessive discharging | | N |
| | Unintentional charging of a non-rechargeable battery | | N |
| | Reverse charging of a rechargeable battery | | N |
| M.3.3 | Compliance criteria | (See appended table M.3) | N |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |

| | | | |
|---|---|---|---|
| **M.4** | **Additional safeguards for equipment containing a secondary lithium battery** | | N |
| M.4.1 | General | | N |
| | IEC 62133-2 batteries used for sub-system power powering application .....: | | N |
| M.4.2 | Charging safeguards | | N |
| M.4.2.1 | Requirements | | N |
| M.4.2.2 | Test | | N |
| M.4.2.2.1 | General | | N |
| M.4.2.2.2 | Abnormal operating conditions | | N |
| M.4.2.2.3 | Single fault conditions | | N |
| M.4.2.3 | Compliance criteria .....: | | N |
| M.4.3 | Fire enclosure .....: | | N |
| M.4.4 | Drop test of equipment containing a secondary lithium battery | | N |
| M.4.4.2 | Preparation and procedure for the drop test | | N |
| M.4.4.3 | Drop, Voltage on reference and dropped batteries (V); voltage difference during 24 h period (%): .....: | | N |
| M.4.4.4 | Check of the charge/discharge function | | N |
| M.4.4.5 | Charge / discharge cycle test | | N |
| M.4.4.6 | Compliance criteria | | N |
| **M.5** | **Risk of burn due to short-circuit during carrying** | | N |
| M.5.1 | Requirement | | N |
| M.5.2 | Test method and compliance criteria | | N |
| **M.6** | **Safeguards against short-circuits** | | N |
| M.6.1 | External and internal faults | | N |
| M.6.2 | Compliance criteria | | N |
| **M.7** | **Risk of explosion from lead acid and NiCd batteries** | | N |
| M.7.1 | Ventilation preventing explosive gas concentration | | N |
| | Calculated hydrogen generation rate.....: | | N |
| M.7.2 | Test method and compliance criteria | | N |
| | Minimum air flow rate, $Q$ ($m^3/h$) .....: | | N |
| M.7.3 | Ventilation tests | | N |
| M.7.3.1 | General | | N |
| M.7.3.2 | Ventilation test – alternative 1 | | N |
| | Hydrogen gas concentration (%) .....: | | N |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| M.7.3.3 | Ventilation test – alternative 2 | | N |
| | Obtained hydrogen generation rate ..... : | | N |
| M.7.3.4 | Ventilation test – alternative 3 | | N |
| | Hydrogen gas concentration (%) ..... : | | N |
| M.7.4 | Marking ..... : | | N |
| **M.8** | **Protection against internal ignition from external spark sources of rechargeable batteries with aqueous electrolyte** | | N |
| M.8.1 | General | | N |
| M.8.2 | Test method | | N |
| M.8.2.1 | General | | N |
| M.8.2.2 | Estimation of hypothetical volume $V_Z$ ($m^3/s$)..... : | | — |
| M.8.2.3 | Correction factors..... : | | — |
| M.8.2.4 | Calculation of distance *d* (mm) ..... : | | — |
| **M.9** | **Preventing electrolyte spillage** | | N |
| M.9.1 | Protection from electrolyte spillage | | N |
| M.9.2 | Tray for preventing electrolyte spillage | | N |
| **M.10** | **Instructions to prevent reasonably foreseeable misuse** | | N |
| | Instructional safeguard..... : | | N |
| **N** | **ELECTROCHEMICAL POTENTIALS** | | N |
| | Material(s) used ..... : | | — |
| **O** | **MEASUREMENT OF CREEPAGE DISTANCES AND CLEARANCES** | | N |
| | Value of *X* (mm) ..... : | | — |
| **P** | **SAFEGUARDS AGAINST CONDUCTIVE OBJECTS** | | N |
| **P.1** | **General** | | N |
| **P.2** | **Safeguards against entry or consequences of entry of a foreign object** | | N |
| P.2.1 | General | The equipment is built-in type, the overall compliance shall be investigated in the complete end system/equipment. | N |
| | Location and Dimensions (mm) ..... : | | — |
| P.2.2 | Safeguard requirements | | N |
| | The ES3 and PS3 keep-out volume in Figure P.4 not applicable to transportable equipment | | N |
| | Transportable equipment with metalized plastic parts ..... : | | N |
| P.2.3 | Consequence of entry test..... : | | N |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| **P.3** | **Safeguards against spillage of internal liquids** | | N |
| P.3.1 | General | | N |
| P.3.2 | Determination of spillage consequences | | N |
| P.3.3 | Spillage safeguards | | N |
| P.3.4 | Compliance criteria | | N |
| **P.4** | **Metallized coatings and adhesives securing parts** | | N |
| P.4.1 | General | | N |
| P.4.2 | Tests | | N |
| | Conditioning, $T_C$ (°C)..... : | | — |
| | Duration (weeks)..... : | | — |
| **Q** | **CIRCUITS INTENDED FOR INTERCONNECTION WITH BUILDING WIRING** | | **N** |
| **Q.1** | **Limited power sources** | | N |
| Q.1.1 | Requirements | | N |
| | a) Inherently limited output | | N |
| | b) Impedance limited output | | N |
| | c) Regulating network limited output | | N |
| | d) Overcurrent protective device limited output | | N |
| | e) IC current limiter complying with G.9 | | N |
| Q.1.2 | Test method and compliance criteria..... : | | N |
| | Current rating of overcurrent protective device (A) : | | N |
| **Q.2** | **Test for external circuits – paired conductor cable** | | N |
| | Maximum output current (A) ..... : | | N |
| | Current limiting method..... : | | — |
| **R** | **LIMITED SHORT CIRCUIT TEST** | | N |
| **R.1** | **General** | | N |
| **R.2** | **Test setup** | | N |
| | Overcurrent protective device for test..... : | | — |
| **R.3** | **Test method** | | N |
| | Cord/cable used for test..... : | | — |
| **R.4** | **Compliance criteria** | | N |
| **S** | **TESTS FOR RESISTANCE TO HEAT AND FIRE** | | **N** |
| **S.1** | **Flammability test for fire enclosures and fire barrier materials of equipment where the steady state power does not exceed 4 000 W** | | N |
| | Samples, material ..... : | | — |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| | Wall thickness (mm)....: | | — |
| | Conditioning (°C)....: | | — |
| | Test flame according to IEC 60695-11-5 with conditions as set out | | N |
| | - Material not consumed completely | | N |
| | - Material extinguishes within 30s | | N |
| | - No burning of layer or wrapping tissue | | N |
| **S.2** | **Flammability test for fire enclosure and fire barrier integrity** | | N |
| | Samples, material ....: | | — |
| | Wall thickness (mm)....: | | — |
| | Conditioning (°C)....: | | — |
| | - Material did not show any additional holes for combustible materials | | N |
| | - Cheesecloth did not ignite for top openings | | N |
| **S.3** | **Flammability test for the bottom of a fire enclosure** | | N |
| S.3.1 | Mounting of samples | | N |
| S.3.2 | Test method and compliance criteria | | N |
| | Mounting of samples ....: | | — |
| | Wall thickness (mm)....: | | — |
| | Cheesecloth did not ignite | | N |
| **S.4** | **Flammability classification of materials** | | N |
| **S.5** | **Flammability test for fire enclosure materials of equipment with a steady state power exceeding 4 000 W** | | N |
| | Samples, material ....: | | — |
| | Wall thickness (mm)....: | | — |
| | Conditioning (°C)....: | | — |
| **S.6** | **Grille covering material, cloth, and reticulated foam** | | N |
| | Samples, material ....: | | — |
| | Measured distance from the centre of the fuel tablet (mm)……: | | N |
| **T** | **MECHANICAL STRENGTH TESTS** | | **N** |
| **T.1** | **General** | | N |
| **T.2** | **Steady force test, 10 N ....:** | | N |
| **T.3** | **Steady force test, 30 N ....:** | | N |
| **T.4** | **Steady force test, 100 N ....:** | | N |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| **T.5** | **Steady force test, 250 N .....:** | | N |
| **T.6** | **Enclosure impact test** | | N |
| | Fall test | | N |
| | Swing test | | N |
| **T.7** | **Drop test .....:** | | N |
| **T.8** | **Stress relief test.....:** | | N |
| **T.9** | **Glass Impact Test.....:** | | N |
| **T.10** | **Glass fragmentation test** | | N |
| | Number of particles counted .....: | | N |
| **T.11** | **Test for telescoping or rod antennas** | | N |
| | Torque value (Nm) .....: | | N |
| **U** | **MECHANICAL STRENGTH OF CATHODE RAY TUBES (CRT) AND PROTECTION AGAINST THE EFFECTS OF IMPLOSION** | | **N** |
| **U.1** | **General** | | N |
| | Instructional safeguard **.....:** | | N |
| **U.2** | **Test method and compliance for non-intrinsically protected CRTs** | | N |
| **U.3** | **Protective screen** | | N |
| **V** | **DETERMINATION OF ACCESSIBLE PARTS** | | **N** |
| **V.1** | **Accessible parts of equipment** | | N |
| V.1.1 | General | The equipment is built-in type, the overall compliance shall be investigated in the complete end system/equipment. | N |
| V.1.2 | Surfaces and openings tested with jointed test probes | | N |
| V.1.3 | Openings tested with straight unjointed test probes | | N |
| V.1.4 | Plugs, jacks, connectors tested with blunt probe | | N |
| V.1.5 | Slot openings tested with wedge probe | | N |
| V.1.6 | Terminals tested with rigid test wire | | N |
| **V.2** | **Accessible part criterion** | | N |
| **X** | **ALTERNATIVE METHOD FOR DETERMINING CLEARANCES FOR INSULATION IN CIRCUITS CONNECTED TO AN AC MAINS NOT EXCEEDING 420 V PEAK (300 V RMS)** | | **N** |
| | Clearance.....: | (See appended table X) | N |
| **Y** | **CONSTRUCTION REQUIREMENTS FOR OUTDOOR ENCLOSURES** | | **N** |
| **Y.1** | **General** | | N |
| **Y.2** | **Resistance to UV radiation** | | N |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| **Y.3** | **Resistance to corrosion** | | N |
| Y.3.1 | Metallic parts of outdoor enclosures are resistant to effects of water-borne contaminants by.....: | | N |
| Y.3.2 | Test apparatus | | N |
| Y.3.3 | Water – saturated sulphur dioxide atmosphere | | N |
| Y.3.4 | Test procedure .....: | | N |
| Y.3.5 | Compliance criteria | | N |
| **Y.4** | **Gaskets** | | N |
| Y.4.1 | General | | N |
| Y.4.2 | Gasket tests | | N |
| Y.4.3 | Tensile strength and elongation tests, changes of tensile strength and elongation .....: | | N |
| | Alternative test methods .....: | | N |
| Y.4.4 | Compression test | | N |
| Y.4.5 | Oil resistance, change of swell / shrink (%) .....: | | N |
| Y.4.6 | Securing means | (See Clause P.4) | N |
| **Y.5** | **Protection of equipment within an outdoor enclosure** | | N |
| Y.5.1 | General | | N |
| Y.5.2 | Protection from moisture | | N |
| | Relevant tests of IEC 60529 or Y.5.3 .....: | | N |
| Y.5.3 | Water spray test | | N |
| Y.5.4 | Protection from plants and vermin | | N |
| Y.5.5 | Protection from excessive dust | | N |
| Y.5.5.1 | General | | N |
| | Relevant tests of IEC 60529 or Y.5.5.2 or Y.5.5.3..: | | N |
| Y.5.5.2 | IP5X equipment | | N |
| Y.5.5.3 | IP6X equipment | | N |
| **Y.6** | **Mechanical strength of enclosures** | | N |
| Y.6.1 | General | | N |
| Y.6.2 | Impact test .....: | (See appended table T.6) | N |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |

**ATTACHMENT TO TEST REPORT**

**IEC 62368-1:2023**
**EUROPEAN GROUP DIFFERENCES AND EUROPEAN NATIONAL DIFFERENCES**
(Audio/video, information and communication technology equipment Part 1: Safety requirements)

**Differences according to**.....: EN IEC 62368-1:2024 + A11:2024

**TRF template used** .....: IECEE OD-2020-F2:2024, Ed. 2

**Attachment Form No**. .....: EU_GD_IEC62368_1F

**Attachment Originator** .....: UL Solutions (Demko)

**Master Attachment** .....: 2025-08-29

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| Clause | Requirement + Test | Result - Remark | Verdict |
|---|---|---|---|
| | **CENELEC COMMON MODIFICATIONS** | | P |
| | Clause numbers in the cells that are shaded light grey are clause references in EN IEC 62368-1:2024+A11:2024. All other clause numbers in that column, except for those in the paragraph below, refers to IEC 62368-1:2023.<br><br>Clauses, subclauses, notes, tables, figures and annexes which are additional to those in IEC 62368-1:2023 are prefixed “Z”. | | P |
| | Add the following annexes:<br>Annex ZA (normative) Normative references to international publications with their corresponding European publications<br>Annex ZB (normative) Special national conditions<br>Annex ZC (informative) A-deviations<br>Annex ZD (informative) IEC and CENELEC code designations for flexible cords | | P |

| | | |
|---|---|---|
| **1** | **MODIFICATION to the whole document** | P |

| EN IEC 62368-1 | | | |
|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |

| | | |
|---|---|---|
| | **Delete all the "country" notes in the reference document according to the following list:**<br><br>0.2.1 Note 1 and Note 2; 1 Note 4 and Note 5; 3.3.8.1 Note 2<br>3.3.8.3 Note 1; 4.1.15 Note; 4.7.3 Note 1 and Note 2<br>5.4.2.3.2.2 Table 12 Note c; 5.4.2.3.2.4 Note 1 and Note 3; 5.4.2.3.2.4 Table 13 Note 2<br>5.4.2.5 Note 2; 5.4.5.1 Note; 5.4.10.2.1 Note<br>5.4.10.2.2 Note; 5.4.10.2.3 Note<br>5.5.2.1 Note; 5.5.6 Note; 5.6.4.2.1 Note 2 and Note 3 and Note 4<br>5.6.8 Note 2; 5.7.7.1 Note 1 and Note 2; 8.5.4.2.3 Note<br>10.2.1 Table 39 Note 3 and Note 4 and Note 5; 10.5.3 Note 2; 10.6.1 Note 3<br>F.3.3.4 Note 2; F.3.3.6 Note 3; Y.4.1 Note<br>Y.4.5 Note | P |
| **2** | **Modification to Clause 1** | N |
| 1 | **Add** the following note at the end of Clause 1:<br><br>"NOTE Z1 The use of certain substances in electrical and electronic equipment is restricted within the EU: see<br>Directive 2011/65/EU."<br><br>Add the following paragraph and note after Note 5:<br><br>"This document is a type test standard.<br><br>NOTE Z2 Routine tests of complete equipment, sub-assemblies or components are covered by EN 62911." | N |
| **3** | **Modification to Clause 2** | N |

| EN IEC 62368-1 | | | |
|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| 2 | Add the following references:<br><br>EN 71-1:2014+A1:2018, Safety of toys - Part 1: Mechanical and physical properties<br><br>EN 50332-1:2013, Sound system equipment: Headphones and earphones associated with personal music players - Maximum sound pressure level measurement methodology - Part 1: General method for "one package equipment"<br><br>EN 50332-2:2013, Sound system equipment: Headphones and earphones associated with personal music players - Maximum sound pressure level measurement methodology - Part 2: Matching of sets with headphones if either or both are offered separately, or are offered as one package equipment but with standardised connectors between the two allowing to combine components of different manufacturers or different design<br><br>EN 50332-3:2017, Sound system equipment: headphones and earphones associated with personal music players - Maximum sound pressure level measurement methodology - Part 3: Measurement method for sound dose management<br><br>IEC/TR 62471-2, Photobiological safety of lamps and lamp systems - Part 2: Guidance on manufacturing requirements relating to non-laser optical radiation safety | | N |
| **4** | **Modification to Clause 4** | | N |

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|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| 4.Z1 | Add the following new subclause 4.Z1 after subclause 4.9:<br><br>"For compliance with B.3 and B.4 in circuits connected to an AC mains, protective devices shall be provided, subject to the following:<br><br>- for pluggable equipment type A, the protective devices shall be included as parts of the equipment, with the exception of components in series with the mains input to the equipment such as the supply cord, appliance coupler, r.f.i. filter and switch, for which the building installation shall be regarded as providing protection in accordance with the rating of the wall socket outlet;<br><br>- for pluggable equipment type B or permanently connected equipment, the protection may be the dedicated overcurrent and short-circuit protection in the building installation, provided that the means of protection, for example a fuse or circuit breaker, is fully specified in the installation instructions.<br><br>Where protective devices are required within the equipment, the protective devices within the equipment shall operate before or at the same time the expected building installation protection will operate.<br><br>For earth faults in single-phase equipment, it is not necessary to provide 2 protective devices. It is expected that the building installation will protect against earth faults. This applies also in countries where an IT power distribution system is used." | | N |
| **5** | **Modification to subclause 4.1.9** | | N |
| 4.1.9 | Add the following paragraph at the end of this subclause:<br><br>"Products need to comply with the requirements of this document with appropriate measurement uncertainty.<br><br>NOTE Z1 See also the RED ADCO position on 'Measurement uncertainty in published harmonized standards'." | | N |
| **6** | **Modification to subclause 5.4.9.1** | | N |

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| Clause | Requirement + Test | Result - Remark | Verdict |
| 5.4.9.1 | Add the following note after the 5th paragraph:<br><br>“NOTE Z1 For guidance on the use of high voltage source, see IEC 60060-1, Clause 8 of IEC 60243-1 and IEC 61180.” | | N |
| **7** | **Modification to subclause 5.4.2.3.2.4** | | N |
| 5.4.2.3.2.4 | Add the following at the end of this subclause:<br><br>“The requirement for interconnection with external circuit in a HBES/BACS network is in addition given in EN IEC 63044-3:2018.” | | N |
| **8** | **Modification to subclause 5.6.6.2** | | N |
| 5.6.6.2 | Replace item d) with the following:<br><br>“d) For equipment powered from a DC mains, if the protective current rating of the circuit under test exceeds 25 A, the test current shall be minimum as required in item a), unless the manufacturer specifies a higher value.” | | N |
| **9** | **Modification to subclause 9.3.1** | | N |
| 9.3.1 | Replace the second paragraph with the following:<br><br>“An accessible part that, while in contact with the body, is likely to drop in temperature upon touch can be evaluated under the limits of Annex A of IEC Guide 117:2010 using the test method of 4.5 of IEC Guide 117.” | | N |
| **10** | **Modification to subclause 10.2.1** | | N |
| 10.2.1 | Add the following to [c] and [d] in Table 38:<br><br>“For additional requirements, see 10.5.1.” | | N |
| **11** | **Modification to subclause 10.4.1** | | N |

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| Clause | Requirement + Test | Result - Remark | Verdict |
| 10.4.1 | Replace the second paragraph of 10.4.1 with:<br><br>“Electronic light effect equipment does not have to comply with the requirements of 10.4. However, 114 IEC/TR 62471-2 shall be considered and proper installation instructions shall be provided.”<br><br>Replace the ninth paragraph of 10.4.1 with:<br><br>“The following information shall be provided in the user manual for safe operation and installation. This information shall also be provided for safe operation by a skilled person who may be exposed to Risk Group 3 energy levels.<br><br>Adequate instructions for proper assembly, installation, maintenance and safe use, including clear warnings concerning precautions to avoid possible exposure to hazardous optical radiation; and<br><br>Advice on safe operating procedures and warnings concerning reasonably foreseeable misuse, malfunctions and hazardous failure modes. Where servicing and maintenance procedures are detailed, they shall include explicit instructions on safe procedures to be followed; and<br><br>The marking on the equipment shall be reproduced in the user manual. A yellow background is not required in the user manual.” | | N |
| **12** | **Modification to subclause 10.4.4** | | N |
| 10.4.4 | Replace the last paragraph of 10.4.4 with:<br><br>“Compliance against material degradation from UV radiation is checked by the applicable tests of Annex C.” | | N |
| **13** | **Modification to subclause 10.5.1** | | N |

| EN IEC 62368-1 | | | |
|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| 10.5.1 | Add the following after the first paragraph:<br><br>“For RS1 compliance is checked by measurement under the following conditions:<br><br>In addition to the normal operating conditions, all controls adjustable from the outside of the equipment by hand, by any object such as a tool or a coin, and those internal adjustments or pre-sets which are not locked in a reliable manner, are adjusted so as to give maximum radiation whilst maintaining an intelligible picture for 1 h, at the end of which the measurement is made.<br><br>NOTE Z1 Soldered joints and paint lockings are examples of adequate locking.<br><br>The dose-rate is determined by means of a radiation monitor with an effective area of 10 cm2, at any point at a distance of 10 cm from the outer surface of the equipment.<br><br>Moreover, the measurement shall be made under fault conditions causing an increase of the high-voltage, provided an intelligible picture is maintained for 1 h, at the end of which the measurement is made.<br><br>For RS1, the dose-rate shall not exceed 1 µSv/h taking account of the background level.<br><br>NOTE Z2 These values appear in Directive 2013/59/Euratom of 5 December 2013.” | | N |
| **14** | **Modification to subclause 10.5.3** | | N |
| 10.5.3 | Replace the second paragraph of 10.5.3 with:<br><br>“The amount of radiation is determined by means of a radiation monitor of the ionizing chamber type with an effective area of 1 000 mm2 ± 10 mm2 or by measuring equipment of other types giving equivalent results.” | | N |
| **15** | **Modification to Clause 10** | | N |
| | Replace 10.6 with the following: | | N |
| 10.6 | **Safeguards against acoustic energy sources** | | N |
| 10.6.1 | **General** | | N |

| EN IEC 62368-1 | | | |
|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| 10.6.1.1 | Introduction<br>Safeguard requirements for protection against long-term exposure to excessive sound pressure levels from personal music players closely coupled to the ear are specified below. Requirements for earphones and headphones intended for use with personal music players are also covered.<br><br>A personal music player is a portable equipment intended for use by an ordinary person, that:<br>- is designed to allow the user to listen to audio or audiovisual content / material; and<br><br>- uses a listening device, such as headphones or earphones that can be worn in or on or around the ears; and<br><br>- has a player that can be body worn (of a size suitable to be carried in a clothing pocket) and is intended for the user to walk around with while in continuous use (for example, on a street, in a subway, at an airport, etc.).<br><br>EXAMPLES Portable CD players, MP3 audio players, mobile phones with MP3 type features, PDAs or similar equipment.<br><br>Personal music players shall comply with the requirements of either 10.6.2 or 10.6.3.<br><br>NOTE 1 Protection against acoustic energy sources from telecom applications is referenced to ITU-T P.360.<br><br>NOTE 2 It is the intention of the Committee to allow the alternative methods for now, but to only use the dose measurement method as given in 10.6.5 in future. Therefore, manufacturers are encouraged to implement 10.6.5 as soon as possible. | | N |

| EN IEC 62368-1 | | | |
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| Clause | Requirement + Test | Result - Remark | Verdict |
| | Listening devices sold separately shall comply with the requirements of 10.6.6.<br><br>These requirements are valid for music or video mode only.<br><br>The requirements do not apply to:<br>- professional equipment;<br><br>NOTE 3 Professional equipment is equipment sold through special sales channels. All products sold through normal electronics stores or general public sales channels are considered not to be professional equipment.<br><br>- hearing aid equipment and other devices for assistive listening;<br><br>- the following type of analogue personal music players:<br>- long distance radio receiver (for example, a multiband radio receiver or world band radio receiver, an AM radio receiver), and<br><br>- cassette player/recorder;<br><br>NOTE 4 This exemption has been allowed because this technology is falling out of use and it is expected that within a few years it will no longer exist. This exemption will not be extended to other technologies.<br><br>- a player while connected to an external amplifier that does not allow the user to walk around while in use;<br><br>- hearing protection devices (HPD) that comply with EN 352-8<br><br>For equipment that is clearly designed or intended primarily for use by children, the limits of the relevant toy standards may apply.<br><br>The relevant requirements are given in EN 71-1:2014+A1:2018, 4.20 and the related tests methods and measurement distances apply. | | |
| 10.6.2 | **Classification of devices without the capacity to estimate sound dose** | | N |

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| Clause | Requirement + Test | Result - Remark | Verdict |
| 10.6.2.1 | General<br><br>This standard is transitioning from short-term based (30 s) requirements to long-term based (40 h) requirements. These clauses remain in effect only for devices that do not comply with sound dose estimation as stipulated in EN 50332-3:2017.<br><br>For classifying the acoustic output *L*Aeq,*T*, measurements are based on the A-weighted equivalent sound pressure level over a 30 s period.<br><br>For music where the average sound pressure (long term *L*Aeq,*T*) measured over the duration of the song is lower than the average produced by the programme simulation noise, measurements may be done over the duration of the complete song. In this case, *T* becomes the duration of the song.<br><br>NOTE Classical music, acoustic music and broadcast typically has an average sound pressure (long term *L*Aeq,*T*) which is much lower than the average programme simulation noise. Therefore, if the player is capable to analyse the content and compare it with the programme simulation noise, the warning does not need to be given as long as the average sound pressure of the song does not exceed the required limit.<br><br>For example, if the player is set with the programme simulation noise to 85 dB, but the average music level of the song is only 65 dB, there is no need to give a warning or ask an acknowledgement as long as the average sound level of the song is not above the basic limit of 85 dB. | | N |

| EN IEC 62368-1 | | | |
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| Clause | Requirement + Test | Result - Remark | Verdict |
| 10.6.2.2 | RS1 limits (to be superseded, see 10.6.3.2)<br><br>RS1 is a class 1 acoustic energy source that does not exceed the following:<br><br>- for equipment provided as a package (player with its listening device), and with a proprietary connector between the player and its listening device, or where the combination of player and listening device is known by other means such as setting or automatic detection, the *L*Aeq,*T* acoustic output shall be ≤ 85 dB when playing the fixed “programme simulation noise” described in EN 50332-1:2013;<br><br>- for equipment provided with a standardized connector (for example, a 3,5 mm headphone/earphone jack) that allows connection to a listening device for general use, the unweighted r.m.s. output voltage shall be ≤ 27 mV (analogue interface) or −25 dBFS (digital interface) when playing the fixed 214 “programme simulation noise” described in EN 50332-1:2013.<br><br>The RS1 limits will be updated for all devices as per 10.6.3.2. | | N |
| 10.6.2.3 | RS2 limits (to be superseded, see 10.6.3.3)<br><br>RS2 is a class 2 acoustic energy source that does not exceed the following:<br><br>- for equipment provided as a package (player with its listening device), and with a proprietary connector between the player and its listening device, or when the combination of player and listening device is known by other means such as setting or automatic detection, the LAeq,T acoustic output shall be ≤ 100 dB(A) when playing the fixed “programme simulation noise” as described in EN 50332-1:2013:<br><br>- for equipment provided with a standardized connector (for example, a 3,5 phone jack) that allows connection to a listening device for general use, the unweighted r.m.s. output voltage shall be ≤ 150 mV (analogue interface) or −10 dBFS (digital interface) when playing the fixed “programme simulation noise” as described in EN 50332-1:2013. | | N |
| 10.6.2.4 | RS3 limits<br><br>RS3 is a class 3 acoustic energy source that exceeds RS2 limits. | | N |

| EN IEC 62368-1 | | | |
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| Clause | Requirement + Test | Result - Remark | Verdict |
| 10.6.3 | **Classification of devices (new)** | | N |
| 10.6.3.1 | General<br><br>Previous limits (10.6.2) created abundant false negative and false positive PMP sound level warnings. New limits, compliant with The Commission Decision 2009/490/EC of 23 June 2009, are given below. | | N |
| 10.6.3.2 | RS1 limits (new)<br><br>RS1 is a class 1 acoustic energy source that does not exceed the following:<br><br>- for equipment provided as a package (player with its listening device), and with a proprietary connector between the player and its listening device, or where the combination of player and listening device is known by other means such as setting or automatic detection, the LAeq,T acoustic output shall be ≤ 80 dB when playing the fixed “programme simulation noise” described in EN 50332-1:2013;<br><br>- for equipment provided with a standardized connector (for example, a 3,5 phone jack) that allows connection to a listening device for general use, the unweighted r.m.s. output voltage shall be ≤ 15 mV (analogue interface) or −30 dBFS (digital interface) when playing the fixed “programme simulation noise” described in EN 50332-1:2013. | | N |

| EN IEC 62368-1 | | | |
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| Clause | Requirement + Test | Result - Remark | Verdict |
| 10.6.3.3 | RS2 limits (new)<br><br>RS2 is a class 2 acoustic energy source that does not exceed the following:<br><br>- for equipment provided as a package (player with its listening device), and with a proprietary connector between the player and its listening device, or where the combination of player and listening device is known by other means such as setting or automatic detection, the weekly sound exposure level, as described in EN 50332-3:2017, shall be ≤ 80 dB when playing the fixed "programme simulation noise" described in 249 EN 50332-1:2013;<br><br>- for equipment provided with a standardized connector (for example, a 3,5 phone jack) that allows connection to a listening device for general use, the unweighted r.m.s. output level, integrated over one week, as described in EN 50332-3:2017, shall be ≤ 15 mV (analogue interface) or −30 dBFS when playing the fixed "programme simulation noise" described in EN 50332‑1:2013. | | N |
| 10.6.4 | **Requirements for maximum sound exposure** | | N |
| 10.6.4.1 | Measurement methods<br><br>All volume controls shall be turned to maximum during tests.<br><br>Measurements shall be made in accordance with EN 50332-1:2013 or EN 50332-2:2013 as applicable. | | N |

| EN IEC 62368-1 | | | |
|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| 10.6.4.2 | Protection of persons<br><br>Except as given below, protection requirements for parts accessible to ordinary persons, instructed persons and skilled persons are given in 4.3.<br><br>NOTE 1 Volume control is not considered to be a safeguard.<br><br>Between RS2 and an ordinary person, the basic safeguard may be replaced by an instructional safeguard in accordance with Clause F.5, except that the instructional safeguard shall be placed on the equipment, or on the packaging, or in the instruction manual. Alternatively, the instructional safeguard may be given through the equipment display during use.<br>The elements of the instructional safeguard shall be as follows:<br><br>- element 1a: the symbol  , IEC 60417-6044 (2011–01)<br>- element 2: “High sound pressure” or equivalent text<br>- element 3: “Hearing damage risk” or equivalent text<br>- element 4: “Do not listen at high volume levels for long periods.” or equivalent text<br><br>An equipment safeguard shall prevent exposure of an ordinary person to an RS2 source without intentional physical action from the ordinary person and shall automatically return to an output level not exceeding what is specified for an RS1 source when the power is switched off.<br><br>The equipment shall provide a means to actively inform the user of the increased sound level when the equipment is operated with an output level exceeding RS1 limits. Any means used shall be acknowledged by the user before activating a mode of operation which allows for an output level exceeding RS1 limits. The acknowledgement does not need to be repeated more than once every 20 h of cumulative listening time. | | N |

| EN IEC 62368-1 | | | |
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| Clause | Requirement + Test | Result - Remark | Verdict |
| | NOTE 2 Examples of means include visual or audible signals. Action from the user is always needed.<br><br>NOTE 3 The 20 h listening time is the accumulative listening time, independent of how often and how long the personal music player has been switched off.<br><br>A skilled person shall not be unintentionally exposed to RS3. | | N |
| 10.6.5 | **Requirements for dose-based systems** | | N |
| 10.6.5.1 | General requirements<br><br>Personal music players shall give the warnings as provided below when tested according to EN 50332-3:2017, using the limits from this clause.<br><br>The manufacturer may offer optional settings to allow the users to modify when and how they wish to receive the notifications and warnings to promote a better user experience without defeating the safeguards. This allows the users to be informed in a method that best meets their physical capabilities and device usage needs. If such optional settings are offered, an administrator (for example, parental restrictions, business/educational administrators, etc.) shall be able to lock any optional settings into a specific configuration.<br><br>The personal music player shall be supplied with easy to understand explanation to the user of the dose management system, the risks involved, and how to use the system safely. The user shall be made aware that other sources may significantly contribute to their sound exposure, for example work, transportation, concerts, clubs, cinema, car races, etc. | | N |

| EN IEC 62368-1 | | | |
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| Clause | Requirement + Test | Result - Remark | Verdict |
| 10.6.5.2 | Dose-based warning and requirements<br><br>When a dose of 100 % CSD is reached, and at least at every 100 % further increase of CSD, the device shall warn the user and require an acknowledgement. In case the user does not acknowledge, the output level shall automatically decrease to a level in compliance with class RS1 limits.<br><br>The warning shall at least clearly indicate that listening above 100 % CSD leads to the risk of hearing damage or loss. | | N |
| 10.6.5.3 | Exposure-based requirements<br><br>With only dose-based requirements, cause and effect could be far separated in time, defying the purpose of educating users about safe listening practice. In addition to dose-based requirements, a PMP shall therefore also put a limit to the short-term sound level a user can listen at.<br><br>The exposure-based limiter (EL) shall automatically reduce the sound level not to exceed 100 dB(A) or 150 mV integrated over the past 180 s, based on methodology defined in EN 50332-3:2017. The EL settling time (time from starting level reduction to reaching target output level) shall be 10 s or less.<br><br>Test of EL functionality is conducted according to EN 50332-3:2017, using the limits from this clause. For equipment provided as a package (player with its listening device), the level integrated over 180 s shall be 100 dB or lower. For equipment provided with a standardized connector, the un-weighted level integrated over 180 s shall be no more than 150 mV for an analogue interface and no more than −10 dBFS for a digital interface.<br><br>In case the source is known not to be music (or test signal), the EL may be disabled. | | N |
| 10.6.6 | **Requirements for listening devices (headphones, earphones, etc.)** | | N |

| EN IEC 62368-1 | | | |
|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| 10.6.6.1 | Corded listening devices with analogue input<br><br>With 94 dB *L*Aeq acoustic pressure output of the listening device, and with the volume and sound settings in the listening device (for example, built-in volume level control, additional sound features like equalization, etc.) set to the combination of positions that maximizes the measured acoustic output level, the input voltage of the listening device when playing the fixed “programme simulation noise” as described in EN 50332-1:2013 shall be ≥ 75 mV.<br><br>NOTE The values of 94 dB and 75 mV correspond with 85 dB and 27 mV in 10.6.2.2. or 100 dB and 150 mV in 10.6.2.3. | | N |
| 10.6.6.2 | Corded listening devices with digital input<br><br>With any playing device playing the fixed “programme simulation noise” described in EN 50332-1:2013, and with the volume and sound settings in the listening device (for example, built-in volume level control, additional sound features like equalization, etc.) set to the combination of positions that maximize the measured acoustic output, the *L*Aeq,*T* acoustic output of the listening device shall be ≤ 100 dB with an input signal of −10 dBFS. | | N |
| 10.6.6.3 | Cordless listening devices<br><br>In cordless mode,<br><br>with any playing and transmitting device playing the fixed programme simulation noise described in EN 50332-1:2013; and<br><br>respecting the cordless transmission standards, where an air interface standard exists that specifies the equivalent acoustic level; and<br><br>with volume and sound settings in the receiving device (for example, built-in volume level control, additional sound features like equalization, etc.) set to the combination of positions that maximize the measured acoustic output for the above mentioned programme simulation noise, the *L*Aeq,*T* acoustic output of the listening device shall be ≤ 100 dB with an input signal of −10 dBFS. | | N |

| EN IEC 62368-1 | | | |
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| Clause | Requirement + Test | Result - Remark | Verdict |
| 10.6.6.4 | Measurement method<br><br>Measurements shall be made in accordance with EN 50332-2:2013 as applicable.” | | N |
| **16** | **Modification to subclause G.3.1.2** | | N |
| G.3.1.2 | Add the following note after the first paragraph:<br><br>“NOTE Z1 An IEC 60730 series standard is considered relevant if the component in question falls within its scope.” | | N |
| **17** | **Modification to subclause G.7.1** | | N |
| G.7.1 | Add the following note at the end of the subclause:<br><br>“NOTE Z1 The harmonized code designations corresponding to the IEC cable types are given in Annex ZD.” | | N |
| **18** | **Modification to subclause M.2** | | N |
| M.2 | Add the following paragraph after the first paragraph:<br><br>“The size of the battery compartment shall be designed taking into account the battery compartment recommendations of the relevant battery standard.<br><br>NOTE For general guidance on the design of the battery compartment, see Clause 8 of IEC 62485-4.” | | N |
| **19** | **Modification to Bibliography** | | N |

| EN IEC 62368-1 | | | |
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| Clause | Requirement + Test | Result - Remark | Verdict |

| | | |
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| | Add the following references:<br><br>EN 60060‑1, High-voltage test techniques – Part 1: General definitions and test requirements<br><br>EN 60898‑1, Electrical accessories – Circuit-breakers for overcurrent protection for household and similar<br><br>installations – Part 1: Circuit-breakers for a.c. operation<br><br>EN 62911, Audio, video and information technology equipment – Routine electrical safety testing in<br><br>production<br><br>ADCO RED Position on 'Measurement uncertainty in published harmonized standards' – https:// ec.europa .eu/ docsroom/ documents/ 32381<br><br>Add the following notes for the standards indicated:<br><br>IEC 60060-1 NOTE Harmonized as EN 60060-1.<br>IEC 60130-9 NOTE Harmonized as EN 60130-9.<br>IEC 60204-1 NOTE Harmonized as EN IEC 60204-1.<br>IEC 60204-11 NOTE Harmonized as EN IEC 60204-11.<br>IEC 60243-1 NOTE Harmonized as EN 60243-1.<br>IEC 60269-2 NOTE Harmonized as HD 60269-2.<br>IEC 60309-1 NOTE Harmonized as EN 60309-1.<br>IEC 60364 NOTE some parts harmonized in HD 384/HD 60364 series.<br>IEC 60601-2-4 NOTE Harmonized as EN 60601-2-4.<br>IEC 60664-5:2005 NOTE Harmonized as EN 60664-5:2007.<br>IEC 60721-3-4 NOTE Harmonized as EN IEC 60721-3-4.<br>IEC 61032:1997 NOTE Harmonized as EN 61032:1998 (not modified).<br>IEC 61180 NOTE Harmonized as EN 61180. | N |

| EN IEC 62368-1 | | | |
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| Clause | Requirement + Test | Result - Remark | Verdict |

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| | IEC 61508-1 NOTE Harmonized as EN 61508-1.<br>IEC 61558-2-1 NOTE Harmonized as EN 61558-2-1.<br>IEC 61558-2-4 NOTE Harmonized as EN 61558-2-4.<br>IEC 61558-2-6 NOTE Harmonized as EN 61558-2-6.<br>IEC 61643-21 NOTE Harmonized as EN 61643-21.<br>IEC 61643-311 NOTE Harmonized as EN 61643-311.<br>IEC 61643-321 NOTE Harmonized as EN 61643-321.<br>IEC 61643-331 NOTE Harmonized as EN IEC 61643-331.<br>IEC 61140:2016 NOTE Harmonized as EN 61140:2016.<br>IEC 61439-5:2014 NOTE Harmonized as EN 61439-5:2015.<br>IEC 61969-3 NOTE Harmonized as EN 61969-3.<br>IEC 62040:2017 NOTE Harmonized as EN IEC 62040:2019.<br>IEC 62305-1 NOTE Harmonized as EN 62305-1.<br>IEC 62368-3 NOTE Harmonized as EN 62368-3.<br>IEC 62485-4 NOTE Harmonized as EN IEC 62485-4.<br>ISO 10218-1 NOTE Harmonized as EN ISO 10218-1.<br>ISO 10218-2 NOTE Harmonized as EN ISO 10218-2.<br>ISO 13482 NOTE Harmonized as EN ISO 13482.<br>ISO 13850 NOTE Harmonized as EN ISO 13850. | | |
| **20** | **Addition of annexes** | | N |
| **ZB** | **ANNEX ZB, SPECIAL NATIONAL CONDITIONS** | | N |
| 4.1.15 | Finland, Norway and Sweden<br><br>To the end of the subclause the following is added:<br><br>Class I pluggable equipment type A intended for connection to other equipment or a network shall, if safety relies on connection to reliable earthing or if surge suppressors are connected between the network terminals and accessible parts, have a marking stating that the equipment shall be connected to an earthed mains socket-outlet.<br><br>The marking text in the applicable countries shall be as follows:<br><br>In Finland: “Laite on liitettävä suojakoskettimilla varustettuun pistorasiaan”<br>In Norway: “Apparatet må tilkoples jordet stikkontakt”<br>In Sweden: “Apparaten skall anslutas till jordat uttag” | | N |

| EN IEC 62368-1 | | | |
|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| 4.7.3 | United Kingdom<br><br>To the end of the subclause the following is added:<br><br>The torque test is performed using a socket-outlet complying with BS 1363, and the plug part shall be assessed to the relevant clauses of BS 1363. Also see Annex G.4.2 of this annex. | | N |
| 5.4.11.1 and Annex G | Finland and Sweden<br><br>To the end of the subclause the following is added:<br><br>For separation of the telecommunication network from earth the following is applicable:<br><br>If this insulation is solid, including insulation forming part of a component, it shall at least consist of either<br><br>-two layers of thin sheet material, each of which shall pass the electric strength test below, or<br><br>-one layer having a distance through insulation of at least 0,4 mm, which shall pass the electric strength test below. | | N |

| EN IEC 62368-1 | | | |
|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| | If this insulation forms part of a semiconductor component (e.g. an optocoupler), there is no distance through insulation requirement for the insulation consisting of an insulating compound completely filling the casing, so that clearances and creepage distances do not exist, if the component passes the electric strength test in accordance with the compliance clause below and in addition<br><br>-passes the tests and inspection criteria of 5.4.7 with an electric strength test of 1,5 kV multiplied by 1,6 (the electric strength test of 5.4.9 shall be performed using 1,5 kV), and<br><br>-is subject to routine testing for electric strength during manufacturing, using a test voltage of 1,5 kV.<br><br>It is permitted to bridge this insulation with a capacitor complying with EN 60384-14, subclass Y2.<br><br>A capacitor classified Y3 according to EN 60384-14, may bridge this insulation under the following conditions:<br><br>-the insulation requirements are satisfied by having a capacitor classified Y3 as defined by EN 60384-14, which in addition to the Y3 testing, is tested with an impulse test of 2,5 kV defined in 5.4.10;<br><br>-the additional testing shall be performed on all the test specimens as described in EN 60384-14;<br><br>-.....the impulse test of 2,5 kV is to be performed before the endurance test in EN 60384-14, in the sequence of tests as described in EN 60384-14. | | |

| EN IEC 62368-1 | | | |
|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| 5.5.2.1 | Norway<br><br>After the 3rd paragraph the following is added:<br><br>Due to the IT power distribution system used, capacitors are required to be rated for the applicable line-to-line voltage (230 V). | | N |
| 5.5.6 | Finland, Norway and Sweden<br><br>To the end of the subclause the following is added:<br><br>Resistors used as basic safeguard or bridging basic insulation in class I pluggable equipment type A shall comply with G.10.1 and the test of G.10.3. | | N |
| 5.6.4.2.1 | Ireland and United Kingdom<br><br>After the indent for pluggable equipment type A, the following is added:<br><br>-.....t<br>he protective current rating is taken to be 13 A, this being the largest rating of fuse used in the mains plug. | | N |
| 5.6.4.2.1 | France<br><br>After the indent for pluggable equipment type A, the following is added:<br>-in certain cases, the protective current rating of the circuit supplied from the mains is taken as 20 A instead of 16 A.<br><br>*Justification:*<br>In France, according to NF C15-100 standard, in certain cases, the maximum rated current of the protective device circuit-breaker is 20 A. | | N |
| 5.6.5.1 | Ireland and United Kingdom<br><br>To the second paragraph the following is added:<br>The range of conductor sizes of flexible cords to be accepted by terminals for equipment with a rated current over 10 A and up to and including 13 A is:<br>1,25 mm2 to 1,5 mm2 in cross-sectional area. | | N |

| EN IEC 62368-1 | | | |
|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| 5.6.8 | Norway<br><br>To the end of the subclause the following is added:<br><br>Equipment connected with an earthed mains plug is classified as class I equipment. See the Norway marking requirement in 4.1.15. The symbol IEC 60417-6092, as specified in F.3.6.2, is accepted. | | N |

| EN IEC 62368-1 | | | |
|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| 5.7.7.1 | Norway and Sweden<br><br>To the end of the subclause the following is added:<br><br>The screen of the television distribution system is normally not earthed at the entrance of the building and there is normally no equipotential bonding system within the building. Therefore the protective earthing of the building installation needs to be isolated from the screen of a cable distribution system.<br><br>It is however accepted to provide the insulation external to the equipment by an adapter or an interconnection cable with galvanic isolator, which may be provided by a retailer, for example.<br><br>The user manual shall then have the following or similar information in Norwegian and Swedish language respectively, depending on in what country the equipment is intended to be used in:<br><br>“Apparatus connected to the protective earthing of the building installation through the mains connection or through other apparatus with a connection to protective earthing – and to a television distribution system using coaxial cable, may in some circumstances create a fire hazard. Connection to a television distribution system therefore has to be provided through a device providing electrical isolation below a certain frequency range (galvanic isolator, see EN 60728-11)”<br><br>NOTE In Norway, due to regulation for CATV-installations, and in Sweden, a galvanic isolator shall provide electrical insulation below 5 MHz. The insulation shall withstand a dielectric strength of 1,5 kV RMS, 50 Hz or 60 Hz, for 1 min.<br><br>Translation to Norwegian (the Swedish text will also be accepted in Norway):<br><br>“Apparater som er koplet til beskyttelsesjord via nettplugg og/eller via annet jordtilkoplet utstyr – og er tilkoplet et koaksialbasert kabel-TV nett, kan forårsake brannfare. For å unngå dette skal det ved tilkopling av apparater til kabel-TV nett installeres en galvanisk isolator mellom apparatet og kabel-TV nettet.” | | N |

| EN IEC 62368-1 | | | |
|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |

| | | | |
|---|---|---|---|
| | Translation to Swedish:<br><br>”Apparater som är kopplad till skyddsjord via jordat vägguttag och/eller via annan utrustning och samtidigt är kopplad till kabel-TV nät kan i vissa fall medföra risk för brand. För att undvika detta skall vid anslutning av apparaten till kabel-TV nät galvanisk isolator finnas mellan apparaten och kabel-TV nätet.”. | | |
| 8.5.4.2.3 | United Kingdom<br><br>Add the following after the 2nd dash bullet in 3rd paragraph:<br><br>An emergency stop system complying with the requirements of EN IEC 60204-1 and EN ISO 13850 is required where there is a risk of personal injury. | | N |
| B.3.1 and B.4 | Ireland and United Kingdom<br><br>The following is applicable:<br><br>To protect against excessive currents and short-circuits in the primary circuit of direct plug-in equipment, tests according to Annexes B.3.1 and B.4 shall be conducted using an external miniature circuit breaker complying with EN 60898-1, Type B, rated 32A. If the equipment does not pass these tests, suitable protective devices shall be included as an integral part of the direct plug-in equipment, until the requirements of Annexes B.3.1 and B.4 are met | | N |
| G.4.2 | United Kingdom<br><br>To the end of the subclause the following is added:<br><br>The plug part of direct plug-in equipment shall be assessed to BS 1363: Part 1, 12.1, 12.2, 12.3, 12.9, 12.11, 12.12, 12.13, 12.16, and 12.17, except that the test of 12.17 is performed at not less than 125 °C. Where the metal earth pin is replaced by an Insulated Shutter Opening Device (ISOD), the requirements of clauses 22.2 and 23 also apply. | | N |

| EN IEC 62368-1 | | | |
|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |
| G.7.1 | United Kingdom<br><br>To the first paragraph the following is added:<br><br>Equipment which is fitted with a flexible cable or cord and is designed to be connected to a mains socket conforming to BS 1363 by means of that flexible cable or cord shall be fitted with a 'standard plug' in accordance with the Plugs and Sockets etc. (Safety) Regulations 1994, Statutory Instrument 1994 No. 1768, unless exempted by those regulations.<br><br>NOTE "Standard plug" is defined in SI 1768:1994 and essentially means an approved plug conforming to BS 1363 or an approved conversion plug. | | N |
| **G.7.1** | Ireland<br><br>To the first paragraph the following is added:<br><br>Apparatus which is fitted with a flexible cable or cord shall be provided with a plug in accordance with Statutory Instrument 525: 1997, "13 A Plugs and Conversion Adapters for Domestic Use Regulations: 1997. S.I. 525 provides for the recognition of a standard of another Member State which is equivalent to the relevant Irish Standard | | N |
| **G.7.2** | Ireland and United Kingdom<br><br>To the first paragraph the following is added:<br><br>A power supply cord with a conductor of 1,25 $mm^2$ is allowed for equipment which is rated over 10 A and up to and including 13 A. | | N |

| EN IEC 62368-1 | | | |
|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |

| | | | |
|---|---|---|---|
| **ZC** | **ANNEX ZC, NATIONAL DEVIATIONS** | | **N** |
| 10.5.2 | Germany<br><br>The following requirement applies:<br><br>For the operation of any cathode ray tube intended for the display of visual images operating at an acceleration voltage exceeding 40 kV, authorization is required, or application of type approval (Bauartzulassung) and marking.<br><br>*Justification*:<br>German ministerial decree against ionizing radiation (Röntgenverordnung), in force since 2002-07-01, implementing the European Directive 96/29/EURATOM.<br><br>NOTE Contact address:<br>Physikalisch-Technische Bundesanstalt, Bundesallee 100, D-38116 Braunschweig,<br>Tel.: Int+49–531–592–6320, Internet: http://www.ptb.de | | N |
| **ZD** | **IEC and CENELEC CODE DESIGNATIONS FOR FLEXIBLE CORDS** | | N |

N

| Type of flexible cord | Code designations | |
|---|---|---|
| | IEC | CENELEC |
| **PVC insulated cords** | | |
| Flat twin tinsel cord | 60227 IEC 41 | H03VH-Y |
| Light polyvinyl chloride sheathed flexible cord | 60227 IEC 52 | H03VV-F<br>H03VVH2-F |
| Ordinary polyvinyl chloride sheathed flexible cord | 60227 IEC 53 | H05VV-F<br>H05VVH2-F |
| **Rubber insulated cords** | | |
| Braided cord | 60245 IEC 51 | H03RT-F |
| Ordinary tough rubber sheathed flexible cord | 60245 IEC 53 | H05RR-F |
| Ordinary polychloroprene sheathed flexible cord | 60245 IEC 57 | H05RN-F |
| Heavy polychloroprene sheathed flexible cord | 60245 IEC 66 | H07RN-F |
| **Cords having high flexibility** | | |
| Rubber insulated and sheathed cord | 60245 IEC 86 | H03RR-H |
| Rubber insulated, crosslinked PVC sheathed cord | 60245 IEC 87 | H03RV4-H |
| Crosslinked PVC insulated and sheathed cord | 60245 IEC 88 | H03V4V4-H |
| **Cords insulated and sheathed with halogen-free thermoplastic compounds** | | |
| Light halogen-free thermoplastic insulated and sheathed flexible cords | | H03Z1Z1-F<br>H03Z1Z1H2-F |
| Ordinary halogen-free thermoplastic insulated and sheathed flexible cords | | H05Z1Z1-F<br>H05Z1Z1H2-F |

| EN IEC 62368-1 | | | |
|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |

| 5.2 | TABLE: Classification of electrical energy sources | | | | | | P |
|---|---|---|---|---|---|---|---|
| Supply Voltage | Location (e.g. circuit designation) | Test conditions | Parameters | | | | ES Class |
| | | | U (V) | I (mA) | Type[1)] | Additional Info [2)] | |
| 3.3Vdc | Input | Normal | 3.3Vdc | -- | SS | -- | ES1 |
| | | Abnormal | 3.3Vdc | -- | SS | -- | |
| | | Single fault-OC/SC | 3.3Vdc | -- | SS | -- | |
| Supplementary information: | | | | | | | |
| 1) Type: Steady state (SS), Capacitance (CP), Single pulse (SP), Repetitive pulses (RP), etc. | | | | | | | |
| 2) Additional Info: Frequency, Pulse duration, Pulse off time, Capacitance value, etc. | | | | | | | |

| 5.4.1.8 | TABLE: Working voltage measurement | | | N |
|---|---|---|---|---|
| Location | RMS voltage (V) | Peak voltage (V) | Frequency (Hz) | Comments |
| -- | -- | -- | -- | -- |
| -- | -- | -- | -- | -- |
| Supplementary information: | | | | |
| | | | | |

| 5.4.1.10.2 | TABLE: Vicat softening temperature of thermoplastics | | | N |
|---|---|---|---|---|
| Method.....: | | ISO 306 / B50 | | -- |
| Object/ Part No./Material | Manufacturer/trademark | Thickness (mm) | T softening (°C) | |
| -- | -- | -- | -- | |
| -- | -- | -- | -- | |
| Supplementary information: | | | | |
| | | | | |

| 5.4.1.10.3 | TABLE: Ball pressure test of thermoplastics | | | | N |
|---|---|---|---|---|---|
| Allowed impression diameter (mm).....: | | ≤ 2 mm | | | -- |
| Object/Part No./Material | Manufacturer/trademark | Thickness (mm) | Test temperature (°C) | Impression diameter (mm) | |
| -- | -- | -- | -- | -- | |
| -- | -- | -- | -- | -- | |

| EN IEC 62368-1 | | | |
|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |

| Supplementary information: |
|---|
| |

| 5.4.2, 5.4.3 | TABLE: Minimum Clearances/Creepage distance | | | | | | | N |
|---|---|---|---|---|---|---|---|---|
| Clearance (cl) and creepage distance (cr) at/of/between: | $U_p$ (V) | $U_{rms}$ (V) | Freq [1] (Hz) | Required cl (mm) | cl (mm) | E.S. [2] (V) | Required cr (mm) | cr (mm) |
| -- | -- | -- | -- | -- | -- | -- | -- | -- |
| -- | -- | -- | -- | -- | -- | -- | -- | -- |
| Supplementary information: | | | | | | | | |
| 1) Only for frequency above 30 kHz<br>2) Complete Electric Strength voltage (E.S. (V) when 5.4.2.4 applied) | | | | | | | | |

| 5.4.4.2 | TABLE: Minimum distance through insulation | | | N |
|---|---|---|---|---|
| Distance through insulation (DTI) at/of | Peak voltage (V) | Insulation | Required DTI (mm) | Measured DTI (mm) |
| -- | -- | -- | -- | -- |
| Supplementary information: | | | | |
| | | | | |

| 5.4.4.9 | TABLE: Solid insulation at frequencies >30 kHz | | | | | N |
|---|---|---|---|---|---|---|
| Insulation material | $E_P$ | Frequency (kHz) | $K_R$ | Thickness $d$ (mm) | Insulation | $V_{PW}$ (Vpk) |
| -- | -- | -- | -- | -- | -- | -- |
| Supplementary information: | | | | | | |
| | | | | | | |

| 5.4.9 | TABLE: Electric strength tests | | | N |
|---|---|---|---|---|
| Test voltage applied between: | Voltage shape (Surge, Impulse, AC, DC, etc.) | Test voltage (V) | Breakdown Yes / No | |
| -- | -- | -- | -- | |
| -- | -- | -- | -- | |
| Supplementary information: | | | | |
| | | | | |

| EN IEC 62368-1 | | | |
|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |

| 5.5.2.2 | TABLE: Stored discharge on capacitors | | | | N |
|---|---|---|---|---|---|
| Location | Supply voltage (V) | Operating and fault condition [1)] | Switch position | Measured voltage (Vpk) | ES Class |
| -- | -- | -- | -- | -- | -- |
| Supplementary information: | | | | | |
| X-capacitors installed for testing:<br>[ ] bleeding resistor rating:<br>[ ] ICX:<br>1) Normal operating condition (e.g., normal operation), SC= short circuit, OC= open circuit | | | | | |

| 5.6.6 | TABLE: Resistance of protective conductors and terminations | | | N |
|---|---|---|---|---|
| Location | Test current (A) | Duration (min) | Voltage drop (V) | Resistance (Ω) |
| -- | -- | -- | -- | -- |
| Supplementary information: | | | | |
| | | | | |

| 5.7.4 | TABLE: Unearthed accessible parts | | | | | N |
|---|---|---|---|---|---|---|
| Location | Operating and fault conditions | Supply Voltage (V) | Parameters | | | ES class |
| | | | Voltage ($V_{rms}$ or $V_{pk}$) | Current ($A_{rms}$ or $A_{pk}$) | Freq. (Hz) | |
| -- | -- | -- | -- | -- | -- | -- |
| Supplementary information: | | | | | | |
| Abbreviation: SC= short circuit; OC= open circuit | | | | | | |

| 5.7.5 | TABLE: Earthed accessible conductive part | | | N |
|---|---|---|---|---|
| Supply voltage (V) ..... : | -- | | | — |
| Phase(s) ..... : | [ ] Single Phase; [ ] Three Phase: [ ] Delta [ ] Wye | | | — |
| Power Distribution System ..... : | ☐ TN ☐ TT ☐ IT | | | — |
| Location | Fault Condition No in IEC 60990 clause 6.2.2 | Touch current (mA) | Comment | |
| -- | -- | -- | -- | |
| -- | -- | -- | -- | |
| Supplementary Information: | | | | |

| EN IEC 62368-1 | | | |
|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |

| |
|---|
| |

| 5.8 | TABLE: Backfeed safeguard in battery backed up supplies | | | | | N |
|---|---|---|---|---|---|---|
| Location | Supply voltage (V) | Operating and fault condition | Time (s) | Open-circuit voltage (V) | Touch current (A) | ES Class |
| -- | -- | -- | -- | -- | -- | -- |
| Supplementary information: | | | | | | |
| Abbreviation: SC= short circuit, OC= open circuit | | | | | | |

| 6.2.2 | TABLE: Power source circuit classifications | | | | | P |
|---|---|---|---|---|---|---|
| Location | Operating and fault condition | Voltage (V) | Current (A) | Max. Power[1)] (W) | Time (S) | PS class |
| 3.3Vdc input | Normal | -- | -- | -- | -- | Consider as PS3 |
| Supplementary information: | | | | | | |
| Abbreviation: SC= short circuit; OC= open circuit | | | | | | |

| 6.2.3.1 | TABLE: Determination of Arcing PIS | | | N |
|---|---|---|---|---|
| Location | Open circuit voltage after 3 s (Vpk) | Measured r.m.s current (A) | Calculated value | Arcing PIS? Yes / No |
| -- | -- | -- | -- | -- |
| Supplementary information: | | | | |
| | | | | |

| 6.2.3.2 | TABLE: Determination of resistive PIS | | P |
|---|---|---|---|
| Location | Operating and fault condition | Dissipate power (W) | Arcing PIS? Yes / No |
| All internal circuits/components | -- | -- | Yes (declaration) |
| Supplementary information: | | | |
| | | | |

| 8.5.5 | TABLE: High pressure lamp | | | N |
|---|---|---|---|---|
| Lamp manufacturer | Lamp type | Explosion method | Longest axis of | Particle found |

| EN IEC 62368-1 | | | |
|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |

| | | | glass particle (mm) | beyond 1 m Yes / No |
|---|---|---|---|---|
| -- | -- | -- | -- | -- |
| Supplementary information: | | | | |
| | | | | |

| 9.6 | TABLE: Temperature measurements for wireless power transmitters | N |
|---|---|---|
| Supply voltage (V)..... | -- | — |
| Max. transmitting power (W)..... | -- | — |

Part A [1)]

| Foreign objects | w/o receiver and direct contact | | with receiver and direct contact | | with receiver and at distance of 2 mm | | with receiver and at distance of 5 mm | |
|---|---|---|---|---|---|---|---|---|
| | Object (°C) | Ambient (°C) | Object (°C) | Ambient (°C) | Object (°C) | Ambient (°C) | Object (°C) | Ambient (°C) |
| Steel disc | -- | -- | -- | -- | -- | -- | -- | -- |
| Aluminium ring | -- | -- | -- | -- | -- | -- | -- | -- |
| Aluminium foil | -- | -- | -- | -- | -- | -- | -- | -- |

| Measurement temperature T of part/at: | w/o receiver and direct contact | | with receiver and direct contact | | with receiver and at distance of 2 mm | | with receiver and at distance of 5 mm | |
|---|---|---|---|---|---|---|---|---|
| | T (°C) | Ambient (°C) | T (°C) | Ambient (°C) | T (°C) | Ambient (°C) | T (°C) | Ambient (°C) |
| -- | -- | -- | -- | -- | -- | -- | -- | -- |
| -- | -- | -- | -- | -- | -- | -- | -- | -- |

| EN IEC 62368-1 | | | |
|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |

| Part B [2)] | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Foreign objects | w/o receiver and direct contact | | with receiver and direct contact | | with receiver and at distance of 2 mm | | with receiver and at distance of 5 mm | |
| | Object (°C) | Ambient (°C) | Object (°C) | Ambient (°C) | Object (°C) | Ambient (°C) | Object (°C) | Ambient (°C) |
| Steel disc | -- | -- | -- | -- | -- | -- | -- | -- |
| Aluminium ring | -- | -- | -- | -- | -- | -- | -- | -- |
| Aluminium foil | -- | -- | -- | -- | -- | -- | -- | -- |
| Measurement temperature T of part/at: | w/o receiver and direct contact | | with receiver and direct contact | | with receiver and at distance of 2 mm | | with receiver and at distance of 5 mm | |
| | T (°C) | Ambient (°C) | T (°C) | Ambient (°C) | T (°C) | Ambient (°C) | T (°C) | Ambient (°C) |
| -- | -- | -- | -- | -- | -- | -- | -- | -- |
| -- | -- | -- | -- | -- | -- | -- | -- | -- |

Supplementary information:

1) The test is performed by powering up the transmitter and then placing each of the foreign objects specified in 9.6.2 in direct contact with the transmitter.

2) The test is performed by first placing each of the foreign objects specified in 9.6.2 in direct contact with the transmitter and then powering up transmitter.

| **5.4.1.4, 9.3, B.1.5, B.2.6** | **TABLE: Temperature measurements** | | | | | | **P** |
|---|---|---|---|---|---|---|---|
| Supply voltage (V) .....: | 3.3Vdc | | | | | | -- |
| Ambient temperature during test $T_{amb}$ (°C) .....: | 20.9 | Shift to Tma | | | | | -- |
| Maximum measured temperature *T* of part/at: | *T* (°C) | | | | | | Allowed $T_{max}$ (°C) |
| Ambient | 20.9 | 105.0 | | | | | -- |
| PCB near U1 on PCB board of unit | 28.4 | 112.5 | | | | | 130 |
| Metal over PCB | 28.1 | 112.2 | | | | | Ref. |
| Accessible parts surface shift to 25°C | | | | | | | |
| Ambient | 25.0 | | | | | | -- |
| -- | -- | | | | | | -- |
| Temperature T of winding: | $t_1$ (°C) | $R_1$ (Ω) | $t_2$ (°C) | $R_2$ (Ω) | T (°C) | Allowed $T_{max}$ (°C) | Insulation class |

| EN IEC 62368-1 | | | |
|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| -- | -- | -- | -- | -- | -- | -- | -- |
| Supplementary information: | | | | | | | |

| B.2.5 | TABLE: Input test | | | | | | | P |
|---|---|---|---|---|---|---|---|---|
| U (V) | Hz | I (A) | I rated (A) | P (W) | P rated (W) | Fuse No | I fuse (A) | Condition/status |
| 3.3Vdc | -- | 0.003 | -- | 0.010 | -- | -- | -- | Max. operating condition. |
| Supplementary information: | | | | | | | | |
| | | | | | | | | |

| B.2.5, E.3.1 | TABLE: Input test for equipment containing audio amplifiers | | | | | | | | | | | N |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Operation Condition: | Signal type | | Frequency (Hz) | | Output loads (Ω) | | Load setup | | | | | |
| -- | -- | | -- | | -- | | -- | | | | | |
| Input | | | | | | | | | Amplifier Output | | | |
| Cond. | U (V) | Hz | I (A) | I rated (A) | P (W) | P rated (W) | Fuse No | I fuse (A) | Ch. | U (V) | P (W) | Load (Ω) |
| -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- |
| Supplementary information: | | | | | | | | | | | | |
| | | | | | | | | | | | | |

| B.3, B.4 | TABLE: Abnormal operating and fault condition tests | | | | | | P |
|---|---|---|---|---|---|---|---|
| Ambient temperature Tamb (°C) ..... : | | | -- | | | | — |
| Power source for EUT: Manufacturer, model/type, outputrating .. : | | | -- | | | | — |
| Component No. | Condition | Supply voltage (V) | Test time | Fuse no. | Fuse current (A) | Observation | |
| C3 | SC | 3.3Vdc | 10mins | -- | -- | Unit operated normally, no high temperature, no hazard, No damaged. | |
| C4 | SC | 3.3Vdc | 10mins | -- | -- | Unit operated normally, no high temperature, no hazard, No damaged. | |
| C8 | SC | 3.3Vdc | 10mins | -- | -- | Unit operated normally, no high temperature, no hazard, | |

| EN IEC 62368-1 | | | |
|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | | | No damaged. |
| Supplementary information: | | | | | | |
| SC: Short-circuited; OC: Open-circuited; OL: Overloaded; BL: Blocked. NHT: No High Temperature; NCD: No Component Damage; NFG: No Flammability Gas. | | | | | | |

| **M.3** | **TABLE: Protection circuits for batteries provided within the equipment** | | | | | | **N** |
|---|---|---|---|---|---|---|---|
| Is it possible to install the battery in a reverse polarity position? ..... : | | | | | — | | — |
| Equipment Specification | Charging | | | | | | |
| | Voltage (V) | | | | Current (A) | | |
| | -- | | | | -- | | |
| Manufacturer/type | Battery specification | | | | | | |
| | Non-rechargeable batteries | | Rechargeable batteries | | | | |
| | Discharging current (A) | Unintentional charging current (A) | Charging | | Discharging current (A) | Reverse charging current (A) | |
| | | | Voltage (V) | Current (A) | | | |
| -- | -- | -- | -- | -- | -- | -- | |
| Note: The tests of M.3.2 are applicable only when above appropriate data is not available. | | | | | | | |
| Specified battery temperature (°C) ..... : | | | | | — | | — |

| Component No. | Fault condition | Charge/ discharge mode | Test time | Temp. (°C) | Current (A) | Voltage (V) | Observation |
|---|---|---|---|---|---|---|---|
| -- | -- | -- | -- | -- | -- | -- | -- |
| -- | -- | -- | -- | -- | -- | -- | -- |
| Supplementary information: | | | | | | | |
| Abbreviation: SC= short circuit; OC= open circuit NL= no chemical leakage; NS= no spillage of liquid; NE= no explosion; NF= no emission of flame or expulsion of molten metal. | | | | | | | |

| **M.4.2** | **TABLE: Charging safeguards for equipment containing a secondary lithium battery** | | N |
|---|---|---|---|
| Maximum specified charging voltage (V) ..... : | | -- | — |
| Maximum specified charging current (A) ..... : | | -- | — |
| Highest specified charging temperature (°C) ..... : | | -- | — |
| Lowest specified charging temperature (°C) ..... : | | -- | — |

| Battery manufacturer/type | Operating and fault condition | Measurement | | | Observation |
|---|---|---|---|---|---|
| | | Charging voltage (V) | Charging current (A) | Temp. (°C) | |

| EN IEC 62368-1 | | | |
|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |

| | | | | | | |
|---|---|---|---|---|---|---|
| -- | -- | -- | -- | -- | -- | -- |
| | -- | -- | -- | -- | -- | -- |
| Supplementary information: | | | | | | |
| Abbreviation: SC= short circuit; OC= open circuit; MSCV= maximum specified charging voltage; MSCC= maximum specified charging current; HSCT= highest specified charging temperature; LSCT= lowest specified charging temperature | | | | | | |

| **Q.1** | **TABLE: Circuits intended for interconnection with building wiring (LPS)** | | | | | | **N** |
|---|---|---|---|---|---|---|---|
| Output Circuit | Condition | $U_{oc}$ (V) | Time (s) | $I_{sc}$ (A) | | S (VA) | |
| | | | | Meas. | Limit | Meas. | Limit |
| -- | -- | -- | -- | -- | -- | -- | -- |
| -- | -- | -- | -- | -- | -- | -- | -- |
| Supplementary Information: | | | | | | | |
| | | | | | | | |

| **T.2, T.3, T.4, T.5** | **TABLE: Steady force test** | | | | | **N** |
|---|---|---|---|---|---|---|
| Part/Location | Material | Thickness (mm) | Probe | Force (N) | Test Duration (s) | Observation |
| -- | -- | -- | -- | -- | -- | -- |
| -- | -- | -- | -- | -- | -- | -- |
| Supplementary information: | | | | | | |
| | | | | | | |

| **T.6, T.9** | **TABLE: Impact test** | | | | **N** |
|---|---|---|---|---|---|
| Location/part | Material | Thickness (mm) | Height (mm) | Observation | |
| -- | -- | -- | -- | -- | |
| -- | -- | -- | -- | -- | |
| Supplementary information: | | | | | |
| | | | | | |

| **T.7** | **TABLE: Drop test** | **N** |
|---|---|---|

| EN IEC 62368-1 | | | |
|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |

| Location/part | Material | Thickness (mm) | Height (mm) | Observation |
|---|---|---|---|---|
| -- | -- | -- | -- | -- |
| -- | -- | -- | -- | -- |
| Supplementary information: | | | | |
| | | | | |

| T.8 | TABLE: Stress relief test | | | | N |
|---|---|---|---|---|---|
| Location/Part | Material | Thickness (mm) | Oven Temperature (°C) | Duration (h) | Observation |
| -- | -- | -- | -- | -- | -- |
| Supplementary information: | | | | | |
| | | | | | |

| X | TABLE: Alternative method for determining minimum clearances distances | | N |
|---|---|---|---|
| Clearance distanced between: | Peak of working voltage (V) | Required cl (mm) | Measured cl (mm) |
| -- | -- | -- | -- |
| Supplementary information: | | | |
| | | | |

| EN IEC 62368-1 | | | |
|---|---|---|---|
| Clause | Requirement + Test | Result - Remark | Verdict |

| 4.1.2 | TABLE: Critical components information | | | | P |
|---|---|---|---|---|---|
| Object / part No. | Manufacturer/ trademark | Type / model | Technical data | Standard | Mark(s) of conformity[1)] |
| PCB | XIAMEN XINANTAIFA INDUSTRIAL CO LTD | ATF-05 | V-0, 130°C | UL 94 UL 796 | UL E201516 |
| (Alt.) | Interchangeable | Interchangeable | V-0, 130°C | UL 94 UL 796 | UL |

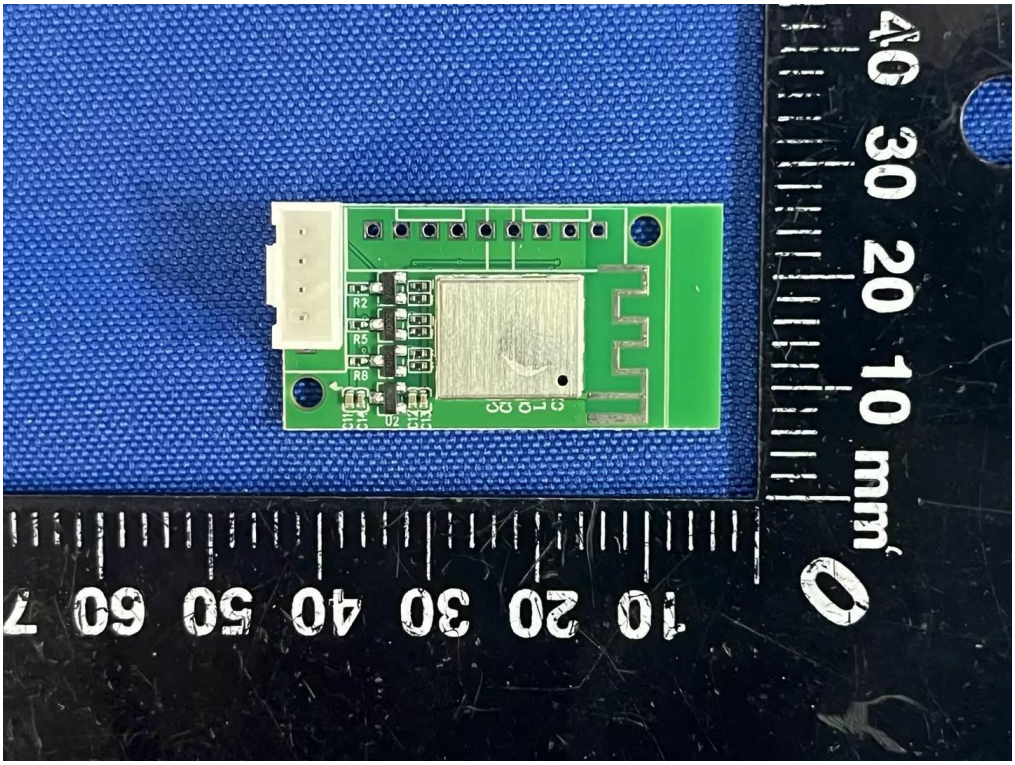
Supplementary information:

[1)] Provided evidence ensures the agreed level of compliance. See OD-2039.

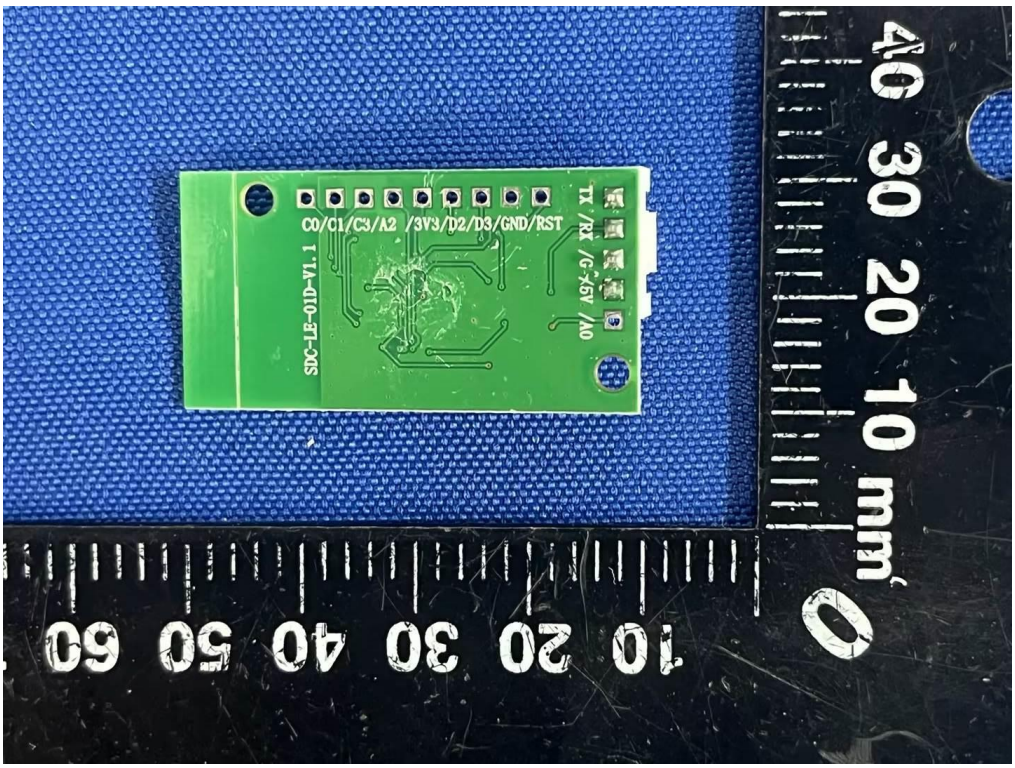
[2)] License available upon request.

# Appendix A - EUT PHOTOS

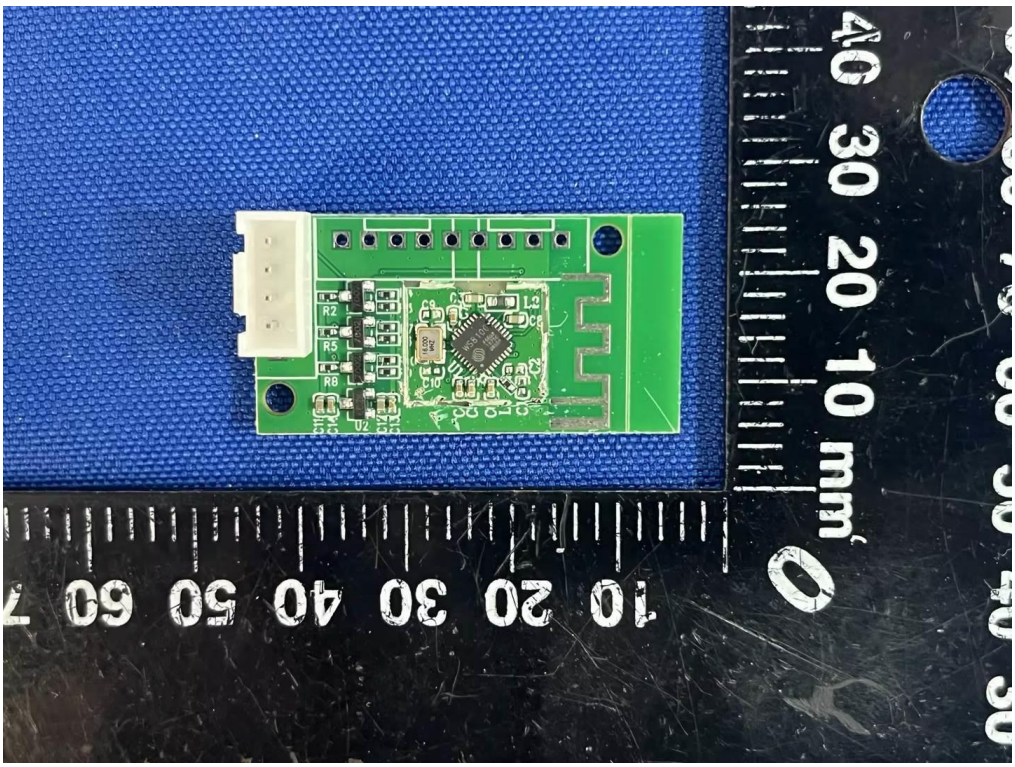
## A.1 EUT- Top View of unit (Model: SDC-LE-01D-4pin)



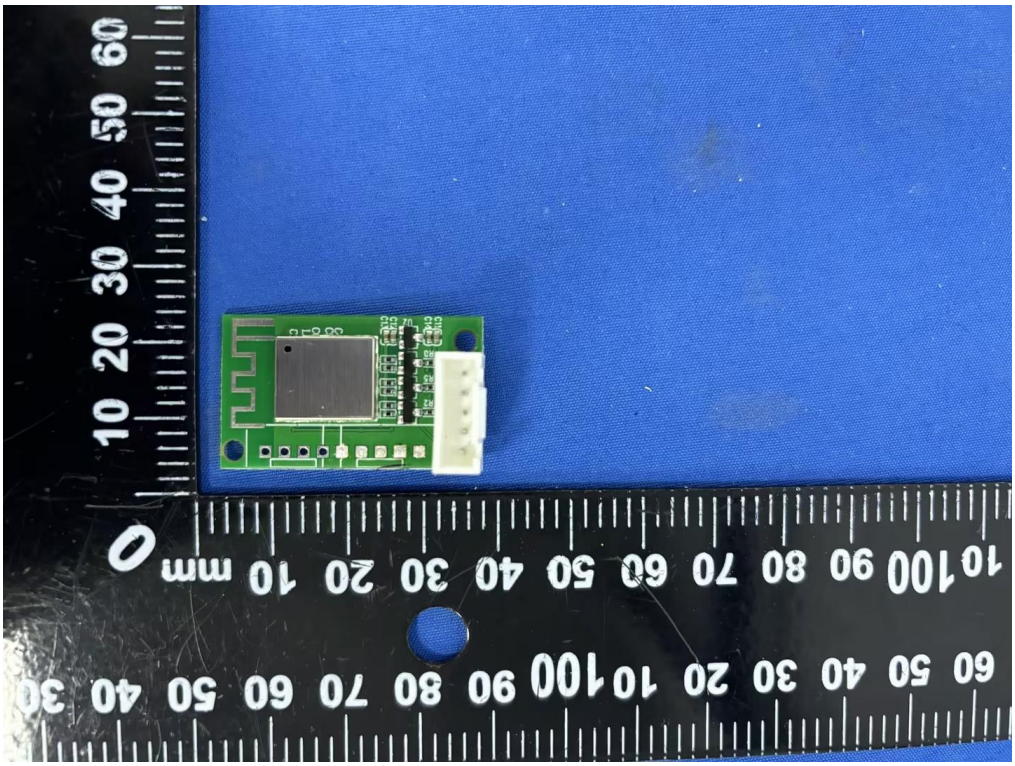
## A.2 EUT- Bottom view of unit (Model: SDC-LE-01D-4pin)



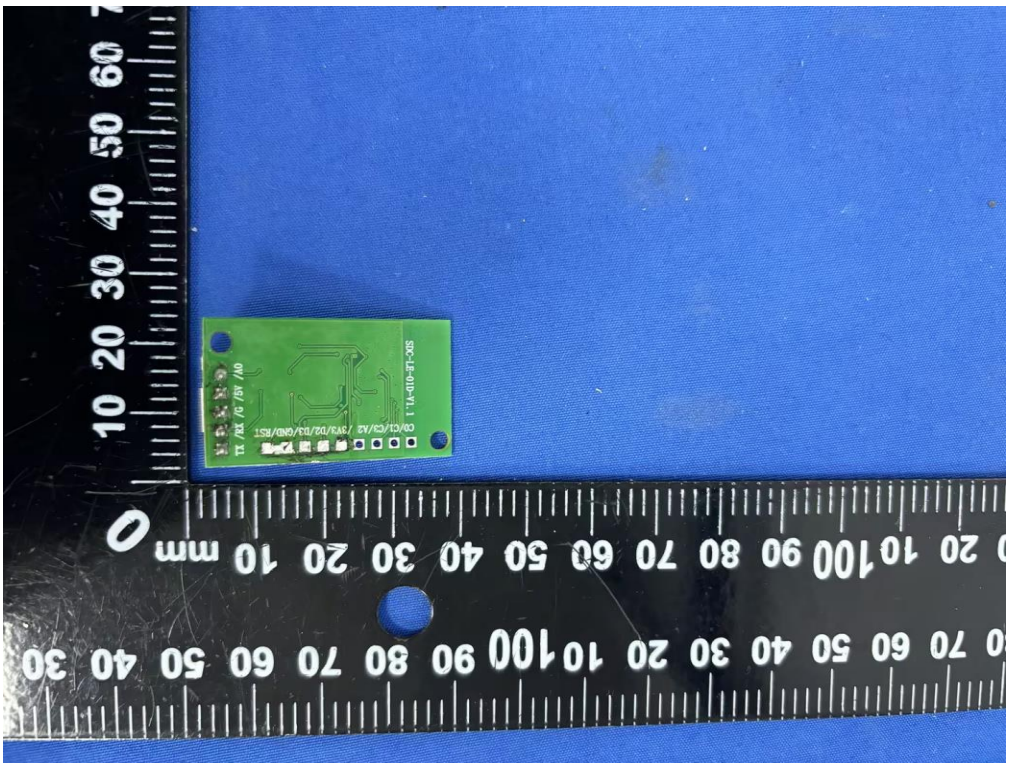
**A.3 EUT- Internal view of unit (Model: SDC-LE-01D-4pin)**



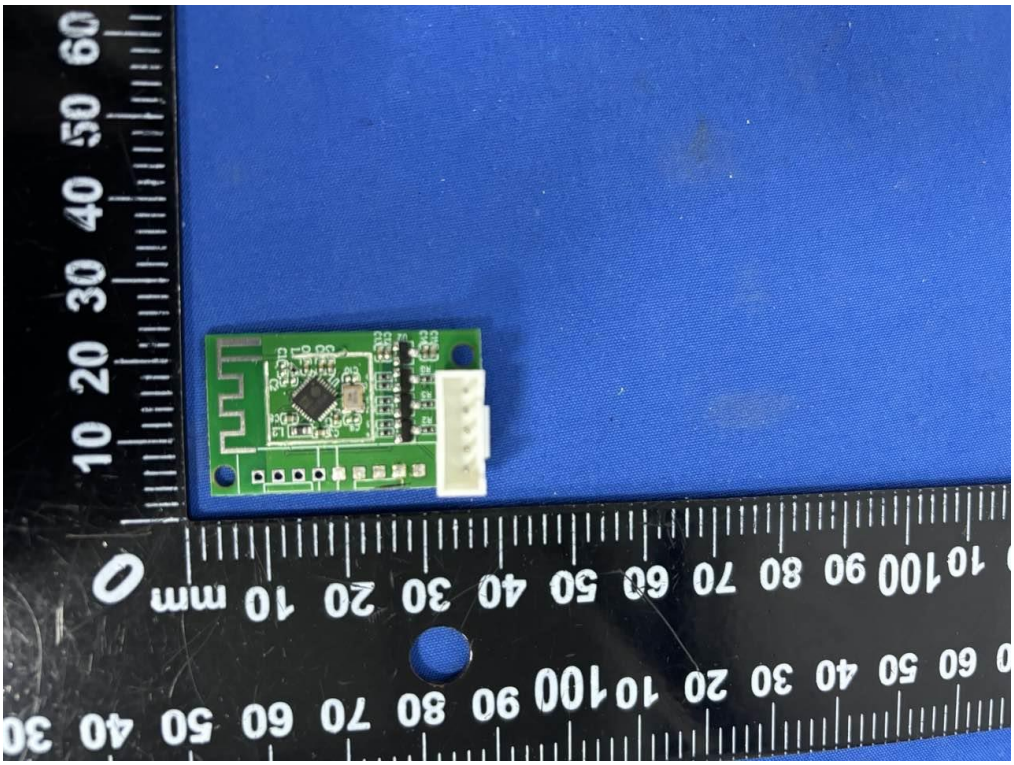
**A.4 EUT- Top View of unit (Model: SDC-LE-01D-5pin)**



**A.5 EUT- Bottom view of unit (Model: SDC-LE-01D-5pin)**



**A.6 EUT- Internal view of unit (Model: SDC-LE-01D-5pin)**



## Appendix B - Declarations

1. The laboratory is not responsible for the authenticity of any information provided by the applicant. Information from the applicant that may affect test results is marked with *.
2. The test data was only valid for the test sample(s).This report must not be duplicated or used in part without prior written consent from the laboratory.
3. Otherwise required by the applicant or Product Regulations, Decision Rule in this report did not consider the uncertainty.
4. This report is valid only with a valid digital signature. The digital signature may be available only under the Adobe software above version 7.0.

*****END OF REPORT*****